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Thermodynamics

JEE Main Chemistry · Previous Year Questions with Answer Key

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Numerical	103 questions
MCQ (Single Correct Answer)	141 questions
Answer key	at the end

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Numerical 103 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Consider the reaction

$2H_2S(g) + 3O_2(g) \rightarrow 2H_2O(l) + 2SO_2(g)$

The magnitude of enthalpy change for the reaction in $kJmol^{-1}$ is ____ . (Nearest integer)

Given : $\Delta_f H^{\ominus}(H_2S) = -20.1\ kJ\ mol^{-1}$
$\Delta_f H^{\ominus}(H_2O) = -286.0\ kJ\ mol^{-1}$
$\Delta_f H^{\ominus}(SO_2) = -297.0\ kJ\ mol^{-1}$

Open solution

Q2. JEE Main 2026 (Online) 6th April Morning Shift

Consider the reaction $X \rightleftharpoons Y$ at 300 K . If ΔH^{θ} and K are $28.40\ kJ\ mol^{-1}$ and 1.8×10^{-7} at the same temperature, then the magnitude of ΔS^{θ} for the reaction in $JK^{-1}\ mol^{-1}$ is ____ . (Nearest integer)

(Given : $R = 8.3\ J\ K^{-1}\ mol^{-1}, \ln 10 = 2.3, \log 3 = 0.48, \log 2 = 0.30$)

Open solution

Q3. JEE Main 2026 (Online) 4th April Evening Shift

If 3.365 g of ethanol (l) is burnt completely in a bomb calorimeter at 298.15 K , the heat produced is 99.472 kJ . The $|\Delta H_f^{\circ}|$ of ethanol at 298.15 K is

____ $\times 10^2\ kJ\ mol^{-1}$. (Nearest integer)

Given: Standard enthalpy for combustion of graphite $= -393.5\ kJ\ mol^{-1}$

Standard enthalpy of formation of water (l) $= -285.8\ kJ\ mol^{-1}$

Molar mass in $gmol^{-1}$ of C, H, O are 12,1 and 16 respectively

Open solution

Q4. JEE Main 2026 (Online) 2nd April Morning Shift

At the transition temperature T, $A \rightleftharpoons B$ and $\Delta G^0 = 105 - 35 \log T$ where A and B are two states of substance X. The transition temperature in $^{\circ}C$ when pressure is 1 atm is _____. (Nearest integer)

Open solution

Q5. JEE Main 2026 (Online) 22nd January Evening Shift

If the enthalpy of sublimation of Li is $155\ kJ\ mol^{-1}$, enthalpy of dissociation of F_2 is $150\ kJ\ mol^{-1}$, ionization enthalpy of Li is $520\ kJ\ mol^{-1}$, electron gain enthalpy of F is $-313\ kJ\ mol^{-1}$, standard enthalpy of formation of LiF is $-594\ kJ\ mol^{-1}$. The magnitude of lattice enthalpy of LiF is ____ $kJmol^{-1}$. (Nearest Integer)

Open solution

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Q6. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Use the following data :

Substance	$\frac{\Delta_f H^{\ominus}(500\ K)}{kJ\ mol^{-1}}$	$\frac{S^{\ominus}(500\ K)}{JK^{-1}\ mol^{-1}}$
$AB(g)$	32	222
$A_2(g)$	6	146
$B_2(g)$	x	280

One mole each of $A_2(g)$ and $B_2(g)$ are taken in a 1 L closed flask and allowed to establish the equilibrium at 500 K .

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

The value of $x\ (\text{in kJ mol}^{-1})$ is ____ . (Nearest integer)

(Given : $\log K = 2.2 \quad R = 8.3\ J\ K^{-1}\ mol^{-1}$)

Q7. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Resonance in X_2Y can be represented as



The enthalpy of formation of X_2Y $\left(X \equiv X(g) + \frac{1}{2}Y = Y(g) \rightarrow X_2Y(g)\right)$ is 80 kJ mol^{-1}. The magnitude of resonance energy of X_2Y is _____ kJ mol^{-1} (nearest integer value).

Given: Bond energies of $X \equiv X$, $X = X$, $Y = Y$ and $X = Y$ are 940, 410, 500, and 602 kJ mol^{-1} respectively.

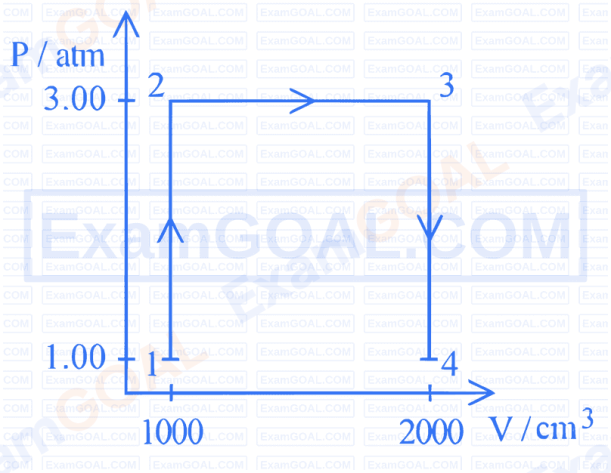
valence X: 3, Y: 2

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Q8. JEE Main 2025 (Online) 3rd April Evening Shift

A perfect gas (0.1 mol) having $\overline{C}_v = 1.50R$ (independent of temperature) undergoes the above transformation from point 1 to point 4. If each step is reversible, the total work done (w) while going from point 1 to point 4 is $(-)$_____J (nearest integer)

[Given: $R = 0.082\ L\ atm\ K^{-1}\ mol^{-1}$]



Open solution

Q9. JEE Main 2025 (Online) 3rd April Evening Shift

A sample of n -octane $(1.14\ g)$ was completely burnt in excess of oxygen in a bomb calorimeter, whose heat capacity is $5\ kJ\ K^{-1}$. As a result of combustion reaction, the temperature of the calorimeter is increased by 5 K . The magnitude of the heat of combustion of octane at constant volume is_____ $kJmol^{-1}$ (nearest integer).

Open solution

Q10. JEE Main 2025 (Online) 3rd April Morning Shift

Given :

$\Delta H^{\ominus}_{sub}[C\ (graphite)] = 710\ kJ\ mol^{-1}$

$\Delta_{C-H}H^{\ominus} = 414\ kJ\ mol^{-1}$

$\Delta_{H-H}H^{\ominus} = 436\ kJ\ mol^{-1}$

$\Delta_{C=C}H^{\ominus} = 611\ kJ\ mol^{-1}$

The $\Delta H^{\ominus}_f$ for $CH_2 = CH_2$ is_____ $kJmol^{-1}$ (nearest integer value)

Open solution

Q11. JEE Main 2025 (Online) 28th January Evening Shift

Consider the following data :

Heat of formation of $CO_2(g) = -393.5\ kJ\ mol^{-1}$

Heat of formation of $H_2O(l) = -286.0\ kJ\ mol^{-1}$

Heat of combustion of benzene $= -3267.0\ kJ\ mol^{-1}$

The heat of formation of benzene is _____ $kJmol^{-1}$. (Nearest integer)

Open solution

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Q12. JEE Main 2025 (Online) 28th January Morning Shift

The formation enthalpies, $\Delta H_f^{\ominus}$ for $H_{(g)}$ and $O_{(g)}$ are 220.0 and $250.0\ kJ\ mol^{-1}$, respectively, at 298.15 K , and $\Delta H_f^{\ominus}$ for $H_2O_{(g)}$ is $-242.0\ kJ\ mol^{-1}$ at the same temperature. The average bond enthalpy of the $O-H$ bond in water at 298.15 K is _____ $kJ\ mol^{-1}$ (nearest integer).

Open solution

Q13. JEE Main 2025 (Online) 24th January Morning Shift

Standard entropies of X_2, Y_2 and XY_5 are 70, 50 and $110\ J\ K^{-1}\ mol^{-1}$ respectively. The temperature in Kelvin at which the reaction

$\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightleftharpoons XY_5 \Delta H^{\ominus} = -35\ kJ\ mol^{-1}$

will be at equilibrium is _____ (Nearest integer)

Open solution

Q14. JEE Main 2025 (Online) 23rd January Evening Shift

The bond dissociation enthalpy of $X_2 \Delta H^{\circ}_{bond}$ calculated from the given data is _____ $kJmol^{-1}$. (Nearest integer)

$M^+X^-(s) \rightarrow M^+(g) + X^-(g) \Delta H^{\circ}_{lattice} = 800\ kJ\ mol^{-1}$

$M(s) \rightarrow M(g) \Delta H^{\circ}_{sub} = 100\ kJ\ mol^{-1}$

$M(g) \rightarrow M^+(g) + e^-(g) \Delta H^{\circ}_{i} = 500\ kJ\ mol^{-1}$

$X(g) + e^-(g) \rightarrow X^-(g) \Delta H^{\circ}_{eg} = -300\ kJ\ mol^{-1}$

$M(s) + \frac{1}{2}X_2(g) \rightarrow M^+X^-(s) \Delta H^{\circ}_{f} = -400\ kJ\ mol^{-1}$

[Given : M^+X^- is a pure ionic compound and X forms a diatomic molecule X_2 in gaseous state]

Open solution

Q15. JEE Main 2025 (Online) 23rd January Morning Shift

The standard enthalpy and standard entropy of decomposition of N_2O_4 to NO_2 are $55.0\ kJ\ mol^{-1}$ and $175.0\ J/K/mol$ respectively. The standard free energy change for this reaction at $25^{\circ}C$ in J mol^{-1} is _____ (Nearest integer)

Open solution

Q16. JEE Main 2025 (Online) 22nd January Evening Shift

Consider the following cases of standard enthalpy of reaction $(\Delta H^{\circ}_r$ in $kJmol^{-1})$

$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \Delta H^{\circ}_1 = -1550$

$C(\text{graphite}) + O_2(g) \rightarrow CO_2(g) \Delta H^{\circ}_2 = -393.5$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \Delta H^{\circ}_3 = -286$

The magnitude of $\Delta H^{\circ}_{fC_2H_6(g)}$ is _____ $kJmol^{-1}$ (Nearest integer).

Open solution

Q17. JEE Main 2024 (Online) 9th April Evening Shift

When $\Delta H_{vap} = 30\ kJ/mol$ and $\Delta S_{vap} = 75\ J\ mol^{-1}\ K^{-1}$, then the temperature of vapour, at one atmosphere is _____ K.

Open solution

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Q18. JEE Main 2024 (Online) 9th April Morning Shift

When equal volume of $1\ \mathrm{M}\ \mathrm{HCl}$ and $1\ \mathrm{M}\ \mathrm{H_2SO_4}$ are separately neutralised by excess volume of $1\mathrm{M}$ NaOH solution. x and $y\ \mathrm{kJ}$ of heat is liberated respectively. The value of y/x is _____.

Open solution

Q19. JEE Main 2024 (Online) 9th April Morning Shift

The heat of solution of anhydrous $\mathrm{CuSO_4}$ and $\mathrm{CuSO_4 \cdot 5H_2O}$ are $-70\ \mathrm{kJ\ mol^{-1}}$ and $+12\ \mathrm{kJ\ mol^{-1}}$ respectively.

The heat of hydration of $\mathrm{CuSO_4}$ to $\mathrm{CuSO_4 \cdot 5H_2O}$ is $-x\ \mathrm{kJ}$. The value of x is _____. (nearest integer).

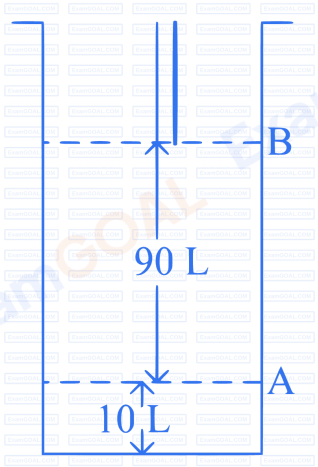
Open solution

Q20. JEE Main 2024 (Online) 8th April Evening Shift

$\Delta_{\text{vap}}\mathrm{H}^{\ominus}$ for water is $+40.79\ \mathrm{kJ\ mol^{-1}}$ at 1 bar and $100^{\circ}\mathrm{C}$. Change in internal energy for this vapourisation under same condition is _____ $\mathrm{kJ\ mol^{-1}}$. (Integer answer) (Given $\mathrm{R} = 8.3\ \mathrm{JK^{-1}\ mol^{-1}}$)

Open solution

Q21. JEE Main 2024 (Online) 8th April Morning Shift



Consider the figure provided.

$1\ \mathrm{mol}$ of an ideal gas is kept in a cylinder, fitted with a piston, at the position A, at $18^{\circ}\mathrm{C}$. If the piston is moved to position B, keeping the temperature unchanged, then 'x' L atm work is done in this reversible process.

x = _____ L atm. (nearest integer)

[Given : Absolute temperature $= {}^{\circ}\mathrm{C} + 273.15, \mathrm{R} = 0.08206\ \mathrm{L\ atm\ mol^{-1}\ K^{-1}}$]

Open solution

Q22. JEE Main 2024 (Online) 6th April Evening Shift

For the reaction at $298\ \mathrm{K}, 2\ \mathrm{A} + \mathrm{B} \rightarrow \mathrm{C}, \Delta\mathrm{H} = 400\ \mathrm{kJ\ mol^{-1}}$ and $\Delta S = 0.2\ \mathrm{kJ\ mol^{-1}\ K^{-1}}$. The reaction will become spontaneous above _____ K.

Open solution

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Q23. JEE Main 2024 (Online) 6th April Morning Shift

An ideal gas, $\overline{C}_v = \frac{5}{2}R$, is expanded adiabatically against a constant pressure of 1 atm until it doubles in volume. If the initial temperature and pressure is $298\ K$ and $5\ atm$, respectively then the final temperature is _____ K (nearest integer).

[$\overline{c}_v$ is the molar heat capacity at constant volume]

Open solution

Q24. JEE Main 2024 (Online) 5th April Evening Shift

Combustion of 1 mole of benzene is expressed as

$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$.

The standard enthalpy of combustion of $2\ mol$ of benzene is $-'x'kJ$. $x =$ _____.

Given :

1. standard Enthalpy of formation of $1\ mol$ of $C_6H_6(l)$, for the reaction $6C$ (graphite) $+3H_2(g) \rightarrow C_6H_6(l)$ is $48.5\ kJ\ mol^{-1}$.

2. Standard Enthalpy of formation of $1\ mol$ of $CO_2(g)$, for the reaction C (graphite) $+O_2(g) \rightarrow CO_2(g)$ is $-393.5\ kJ\ mol^{-1}$.

3. Standard Enthalpy of formation of $1\ mol$ of $H_2O(l)$, for the reaction $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ is $-286\ kJ\ mol^{-1}$.

Open solution

Q25. JEE Main 2024 (Online) 5th April Morning Shift

The heat of combustion of solid benzoic acid at constant volume is $-321.30\ kJ$ at $27^\circ C$. The heat of combustion at constant pressure is $(-321.30 - xR)\ kJ$, the value of x is _____.

Open solution

Q26. JEE Main 2024 (Online) 4th April Evening Shift

Three moles of an ideal gas are compressed isothermally from $60\ L$ to $20\ L$ using constant pressure of $5\ atm$. Heat exchange Q for the compression is - _____ Lit. atm.

Open solution

Q27. JEE Main 2024 (Online) 4th April Morning Shift

The enthalpy of formation of ethane (C_2H_6) from ethylene by addition of hydrogen where the bond-energies of $C-H, C-C, C=C, H-H$ are $414\ kJ, 347\ kJ, 615\ kJ$ and $435\ kJ$ respectively is $-$ _____ kJ

Open solution

Q28. JEE Main 2024 (Online) 1st February Evening Shift

For a certain reaction at $300\ K$, $K = 10$, then ΔG° for the same reaction is - _____ $\times 10^{-1}\ kJ\ mol^{-1}$.

(Given $R = 8.314 JK^{-1}\ mol^{-1}$)

Open solution

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Q29. JEE Main 2024 (Online) 31st January Evening Shift

If 5 moles of an ideal gas expands from $10\,\mathrm{L}$ to a volume of $100\,\mathrm{L}$ at $300\,\mathrm{K}$ under isothermal and reversible condition then work, w, is $-x$ J. The value of x is _____.

(Given R = 8.314 J K^{-1} mol^{-1})

Open solution

Q30. JEE Main 2024 (Online) 31st January Morning Shift

Consider the following reaction at $298\,\mathrm{K}$: $\frac{3}{2}\mathrm{O}_{2(g)} \rightleftharpoons \mathrm{O}_{3(g)}; \mathrm{K_P} = 2.47 \times 10^{-29}$. $\Delta_r G^{\ominus}$ for the reaction is _____ kJ. (Given $\mathrm{R} = 8.314\,\mathrm{JK^{-1}\,mol^{-1}}$)

Open solution

Q31. JEE Main 2024 (Online) 30th January Evening Shift

Two reactions are given below:

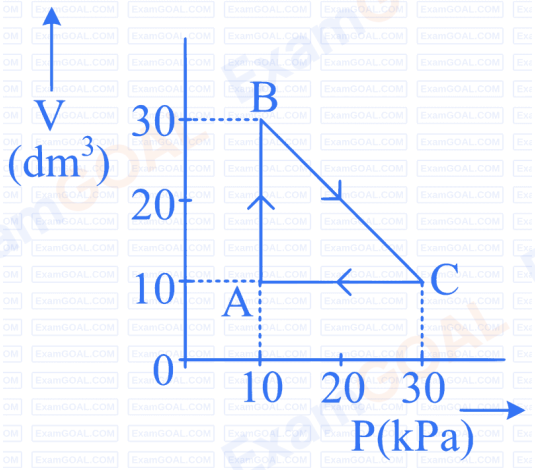
$$2\mathrm{Fe}_{(s)} + \frac{3}{2}\mathrm{O}_{2(g)} \rightarrow \mathrm{Fe_2O_{3(s)}}, \Delta\mathrm{H}^{\circ} = -822\,\mathrm{kJ/mol}$$

$$\mathrm{C}_{(s)} + \frac{1}{2}\mathrm{O}_{2(g)} \rightarrow \mathrm{CO}_{(g)}, \Delta\mathrm{H}^{\circ} = -110\,\mathrm{kJ/mol}$$

Then enthalpy change for following reaction $3\mathrm{C}_{(s)} + \mathrm{Fe_2O_{3(s)}} \rightarrow 2\mathrm{Fe}_{(s)} + 3\mathrm{CO}_{(g)}$ is _____ $\mathrm{kJ/mol}$.

Open solution

Q32. JEE Main 2024 (Online) 30th January Morning Shift



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path $\mathrm{A} \rightarrow \mathrm{B} \rightarrow \mathrm{C} \rightarrow \mathrm{A}$ as shown in the diagram above. The total work done in the process is _____ J.

Open solution

Q33. JEE Main 2024 (Online) 29th January Evening Shift

Standard enthalpy of vapourisation for $\mathrm{CCl_4}$ is $30.5\,\mathrm{kJ\,mol^{-1}}$. Heat required for vapourisation of $284\,\mathrm{g}$ of $\mathrm{CCl_4}$ at constant temperature is _____ kJ.

(Given molar mass in $\mathrm{gmol^{-1}}$; $\mathrm{C} = 12, \mathrm{Cl} = 35.5$)

Open solution

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Q34. JEE Main 2024 (Online) 27th January Evening Shift

For a certain thermochemical reaction $M \rightarrow N$ at $T = 400\ K, \Delta H^{\ominus} = 77.2\ kJ\ mol^{-1}, \Delta S = 122\ JK^{-1}$, log equilibrium constant $(\log K)$ is _____ $\times 10^{-1}$.

Open solution

Q35. JEE Main 2024 (Online) 27th January Morning Shift

If three moles of an ideal gas at $300\ K$ expand isothermally from $30\ dm^3$ to $45\ dm^3$ against a constant opposing pressure of $80\ kPa$, then the amount of heat transferred is _____ J.

Open solution

Q36. JEE Main 2023 (Online) 15th April Morning Shift

$30.4\ kJ$ of heat is required to melt one mole of sodium chloride and the entropy change at the melting point is $28.4\ J\ K^{-1}\ mol^{-1}$ at 1 atm. The melting point of sodium chloride is _____ K (Nearest Integer)

Open solution

Q37. JEE Main 2023 (Online) 13th April Morning Shift

$A_2 + B_2 \rightarrow 2AB.\ \Delta H_f^0 = -200\ kJ\ mol^{-1}$

AB, A_2 and B_2 are diatomic molecules. If the bond enthalpies of A_2, B_2 and AB are in the ratio $1 : 0.5 : 1$, then the bond enthalpy of A_2 is _____ $kJ\ mol^{-1}$ (Nearest integer)

Open solution

Q38. JEE Main 2023 (Online) 12th April Morning Shift

One mole of an ideal gas at $350\ K$ is in a $2.0\ L$ vessel of thermally conducting walls, which are in contact with the surroundings. It undergoes isothermal reversible expansion from 2.0 L to $3.0\ L$ against a constant pressure of $4\ atm$. The change in entropy of the surroundings (ΔS) is _____ JK^{-1} (Nearest integer)

Given: $R = 8.314\ J\ K^{-1}\ mol^{-1}$.

Open solution

Q39. JEE Main 2023 (Online) 11th April Evening Shift

The total number of intensive properties from the following is _____

Volume, Molar heat capacity, Molarity, E^{θ} cell, Gibbs free energy change, Molar mass, Mole

Open solution

Q40. JEE Main 2023 (Online) 11th April Morning Shift

Solid fuel used in rocket is a mixture of Fe_2O_3 and Al (in ratio 1 : 2). The heat evolved (kJ) per gram of the mixture is _____. (Nearest integer)

Given: $\Delta H_f^{\theta}(Al_2O_3) = -1700\ kJ\ mol^{-1}$

$\Delta H_f^{\theta}(Fe_2O_3) = -840\ kJ\ mol^{-1}$

Molar mass of Fe, Al and O are 56, 27 and 16 g mol^{-1} respectively.

Open solution

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Q41. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

The number of endothermic process/es from the following is _____.

A. $I_2(g) \rightarrow 2I(g)$

B. $HCl(g) \rightarrow H(g) + Cl(g)$

C. $H_2O(l) \rightarrow H_2O(g)$

D. $C(s) + O_2(g) \rightarrow CO_2(g)$

E. Dissolution of ammonium chloride in water

Q42. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

For complete combustion of ethene.

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(l)$

the amount of heat produced as measured in bomb calorimeter is $1406\ kJ\ mol^{-1}$ at $300\ K$. The minimum value of $T\Delta S$ needed to reach equilibrium is $(-)$ _____ kJ. (Nearest integer)

Given : $R = 8.3\ J\ K^{-1}\ mol^{-1}$

Q43. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

When a $60\ W$ electric heater is immersed in a gas for 100 s in a constant volume container with adiabatic walls, the temperature of the gas rises by $5^{\circ}C$. The heat capacity of the given gas is _____ JK^{-1} (Nearest integer)

Q44. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Consider the following data

Heat of combustion of $H_2(g)$ $= -241.8\ kJ\ mol^{-1}$

Heat of combustion of $C(s)$ $= -393.5\ kJ\ mol^{-1}$

Heat of combustion of $C_2H_5OH(l)$ $= -1234.7\ kJ\ mol^{-1}$

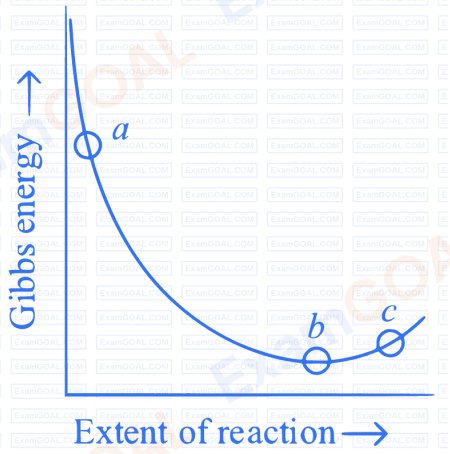
The heat of formation of $C_2H_5OH(l)$ is $(-)$ _____ $kJ\ mol^{-1}$ (Nearest integer).

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Q45. JEE Main 2023 (Online) 6th April Morning Shift

Consider the graph of Gibbs free energy G vs Extent of reaction. The number of statement/s from the following which are true with respect to points (a), (b) and (c) is _____

Open solution



A. Reaction is spontaneous at (a) and (b)

B. Reaction is at equilibrium at point (b) and non-spontaneous at point (c)

C. Reaction is spontaneous at (a) and non-spontaneous at (c)

D. Reaction is non-spontaneous at (a) and (b)

Q46. JEE Main 2023 (Online) 6th April Morning Shift

The value of $\log \mathrm{K}$ for the reaction $\mathrm{A} \rightleftharpoons \mathrm{B}$ at $298\ \mathrm{K}$ is _____. (Nearest integer)

Given: $\Delta \mathrm{H}^{\circ} = -54.07\ \mathrm{kJ\ mol^{-1}}$

$\Delta \mathrm{S}^{\circ} = 10\ \mathrm{J\ K^{-1}\ mol^{-1}}$

(Take $2.303 \times 8.314 \times 298 = 5705$)

Open solution

Q47. JEE Main 2023 (Online) 1st February Evening Shift

$0.3\ \mathrm{g}$ of ethane undergoes combustion at $27^{\circ}\mathrm{C}$ in a bomb calorimeter. The temperature of calorimeter system (including the water) is found to rise by $0.5^{\circ}\mathrm{C}$. The heat evolved during combustion of ethane at constant pressure is _____ $\mathrm{kJ\ mol^{-1}}$. (Nearest integer)

[Given : The heat capacity of the calorimeter system is $20\ \mathrm{kJ\ K^{-1}}, \mathrm{R} = 8.3\ \mathrm{JK^{-1}\ mol^{-1}}$.

Assume ideal gas behaviour.

Atomic mass of C and H are 12 and $1\ \mathrm{g\ mol^{-1}}$ respectively]

Open solution

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Q48. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

At $25^{\circ}C$, the enthalpy of the following processes are given :

$H_2(g) + O_2(g)$	$\rightarrow$	$2OH(g)$	$\Delta H^{\circ} = 78\ kJ\ mol^{-1}$
$H_2(g) + \frac{1}{2}O_2(g)$	$\rightarrow$	$H_2O(g)$	$\Delta H^{\circ} = -242\ kJ\ mol^{-1}$
$H_2(g)$	$\rightarrow$	$2H(g)$	$\Delta H^{\circ} = 436\ kJ\ mol^{-1}$
$\frac{1}{2}O_2(g)$	$\rightarrow$	$O(g)$	$\Delta H^{\circ} = 249\ kJ\ mol^{-1}$

What would be the value of X for the following reaction ? _____ (Nearest integer)

$H_2O(g) \rightarrow H(g) + OH(g)\ \Delta H^{\circ} = X\ kJ\ mol^{-1}$

Q49. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

Enthalpies of formation of $CCl_4(g), H_2O(g), CO_2(g)$ and $HCl(g)$ are $-105, -242, -394$ and $-92\ kJ\ mol^{-1}$ respectively. The magnitude of enthalpy of the reaction given below is _____ $kJ\ mol^{-1}$. (nearest integer)

$CCl_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4HCl(g)$

Q50. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

The enthalpy change for the conversion of $\frac{1}{2}Cl_2(g)$ to Cl^-(aq) is $(-)$ _____ $kJmol^{-1}$ (Nearest integer)

Given : $\Delta_{dis}H^{\ominus}_{Cl^{\theta}_{2(g)}} = 240\ kJ\ mol^{-1}, \Delta_{eg}H^{\ominus}_{Cl_{(g)}} = -350\ kJ\ mol^{-1}$,

$\Delta_{hyd}H^{\Theta}_{Cl^-_{(g)}} = -380\ kJ\ mol^{-1}$

Q51. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

1 mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of $27^{\circ}C$. The work done is $3\ kJ\ mol^{-1}$. The final temperature of the gas is _____ K (Nearest integer).

Given $C_V = 20\ J\ mol^{-1}\ K^{-1}$

Q52. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

When 2 litre of ideal gas expands isothermally into vacuum to a total volume of 6 litre, the change in internal energy is _____ J. (Nearest integer)

Q53. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

28.0 L of CO_2 is produced on complete combustion of 16.8 L gaseous mixture of ethene and methane at 25°C and 1 atm. Heat evolved during the combustion process is _____ kJ.

Given : $\Delta H_c\ (CH_4) = -900\ kJ\ mol^{-1}$

$\Delta H_c\ (C_2H_4) = -1400\ kJ\ mol^{-1}$

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Q54. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

An athlete is given 100 g of glucose ($C_6H_{12}O_6$) for energy. This is equivalent to 1800kJ of energy. The 50% of this energy gained is utilized by the athlete for sports activities at the event. In order to avoid storage of energy, the weight of extra water he would need to perspire is _____ g (Nearest integer)

Assume that there is no other way of consuming stored energy.

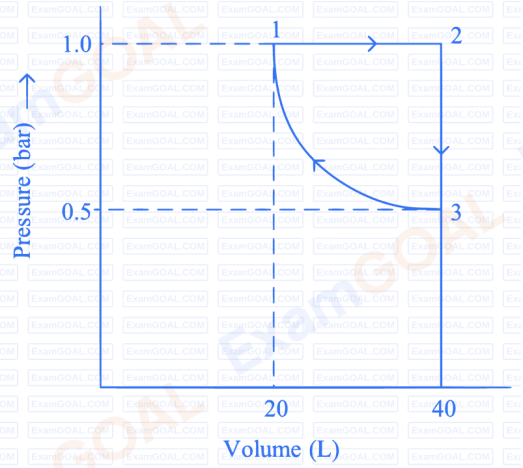
Given : The enthalpy of evaporation of water is 45 kJ mol^{-1}

Molar mass of C, H & O are 12, 1 and 16 g mol^{-1}.

Q55. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

One mole of an ideal monoatomic gas is subjected to changes as shown in the graph. The magnitude of the work done (by the system or on the system) is _____ J (nearest integer)



Given ; $\log 2 = 0.3$

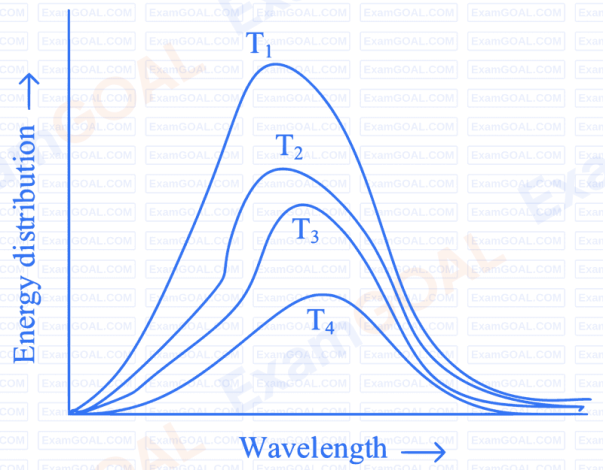
$\ln 10 = 2.3$

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Q56. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

Following figure shows spectrum of an ideal black body at four different temperatures. The number of correct statement/s from the following is _____.



A. $T_4 > T_3 > T_2 > T_1$

B. The black body consists of particles performing simple harmonic motion.

C. The peak of the spectrum shifts to shorter wavelength as temperature increases.

D. $\frac{T_1}{v_1} = \frac{T_2}{v_2} = \frac{T_3}{v_3} \neq$ constant

E. The given spectrum could be explained using quantisation of energy.

Q57. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

For independent processes at 300 K

Process	$\Delta H/kJ\,mol^{-1}$	$\Delta S/J\,K^{-1}$
A	-25	-80
B	-22	40
C	25	-50
D	22	20

The number of non-spontaneous processes from the following is _____

Q58. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

When 600 mL of 0.2 M HNO_3 is mixed with $400\,mL$ of 0.1 M NaOH solution in a flask, the rise in temperature of the flask is _____ $\times 10^{-2}\,^{\circ}C$.

(Enthalpy of neutralisation $= 57\,kJ\,mol^{-1}$ and Specific heat of water $= 4.2\,JK^{-1}\,g^{-1}$) (Neglect heat capacity of flask)

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Q59. JEE Main 2022 (Online) 28th July Evening Shift

Among the following the number of state variables is _____.

Internal energy (U)

Volume (V)

Heat (q)

Enthalpy (H)

Open solution

Q60. JEE Main 2022 (Online) 27th July Evening Shift

A gas (Molar mass = 280 $\mathrm{g\,mol^{-1}}$) was burnt in excess O_2 in a constant volume calorimeter and during combustion the temperature of calorimeter increased from $298.0\ \mathrm{K}$ to $298.45\ \mathrm{K}$. If the heat capacity of calorimeter is $2.5\ \mathrm{kJ\,K^{-1}}$ and enthalpy of combustion of gas is $9\ \mathrm{kJ\,mol^{-1}}$ then amount of gas burnt is _____ g. (Nearest Integer)

Open solution

Q61. JEE Main 2022 (Online) 27th July Morning Shift

The molar heat capacity for an ideal gas at constant pressure is $20.785\ \mathrm{J\,K^{-1}\,mol^{-1}}$. The change in internal energy is $5000\ \mathrm{J}$ upon heating it from $300\ \mathrm{K}$ to $500\ \mathrm{K}$. The number of moles of the gas at constant volume is _____. [Nearest integer] (Given: $\mathrm{R} = 8.314\ \mathrm{J\,K^{-1}\,mol^{-1}}$)

Open solution

Q62. JEE Main 2022 (Online) 26th July Evening Shift

For the reaction

$\mathrm{H_2F_2(g) \rightarrow H_2(g) + F_2(g)}$

$\Delta U = -59.6\ \mathrm{kJ\,mol^{-1}}$ at $27^\circ\mathrm{C}$.

The enthalpy change for the above reaction is $(-)$ _____ $\mathrm{kJ\,mol^{-1}}$ [nearest integer]

Given: $\mathrm{R} = 8.314\ \mathrm{J\,K^{-1}\,mol^{-1}}$.

Open solution

Q63. JEE Main 2022 (Online) 26th July Morning Shift

$2.4\ \mathrm{g}$ coal is burnt in a bomb calorimeter in excess of oxygen at $298\ \mathrm{K}$ and $1\ \mathrm{atm}$ pressure. The temperature of the calorimeter rises from $298\ \mathrm{K}$ to $300\ \mathrm{K}$. The enthalpy change during the combustion of coal is $-x\ \mathrm{kJ\,mol^{-1}}$. The value of x is _____. (Nearest Integer)

(Given : Heat capacity of bomb calorimeter $20.0\ \mathrm{kJ\,K^{-1}}$. Assume coal to be pure carbon)

Open solution

Q64. JEE Main 2022 (Online) 25th July Evening Shift

While performing a thermodynamics experiment, a student made the following observations.

HCl + NaOH $\rightarrow$ NaCl + H_2O $\Delta H = -57.3\ \mathrm{kJ\,mol^{-1}}$

CH_3COOH + NaOH $\rightarrow$ CH_3COONa + H_2O $\Delta H = -55.3\ \mathrm{kJ\,mol^{-1}}$

The enthalpy of ionization of CH_3COOH as calculated by the student is _____ $\mathrm{kJ\,mol^{-1}}$. (nearest integer)

Open solution

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Q65. JEE Main 2022 (Online) 25th July Morning Shift

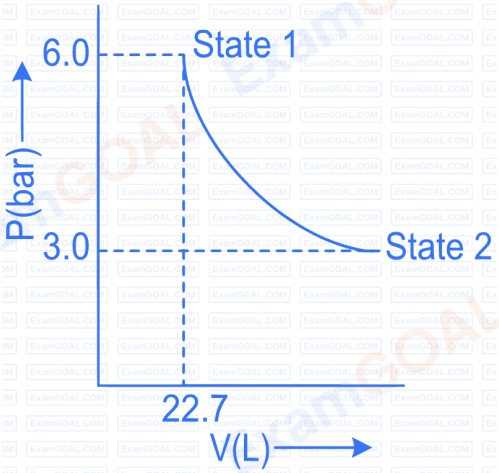
The enthalpy of combustion of propane, graphite and dihydrogen at $298\ \mathrm{K}$ are $-2220.0\ \mathrm{kJ\ mol^{-1}}$, $-393.5\ \mathrm{kJ\ mol^{-1}}$ and $-285.8\ \mathrm{kJ\ mol^{-1}}$ respectively. The magnitude of enthalpy of formation of propane (C_3H_8) is _____ $\mathrm{kJ\ mol^{-1}}$. (Nearest integer)

Open solution

Q66. JEE Main 2022 (Online) 30th June Morning Shift

1.0 mol of monoatomic ideal gas is expanded from state 1 to state 2 as shown in the figure. The magnitude of the work done for the expansion of gas from state 1 to state 2 at 300 K is _____ J. (Nearest integer)

(Given : R = 8.3 J K^{-1} mol^{-1}, ln10 = 2.3, log2 = 0.30)



Open solution

Q67. JEE Main 2022 (Online) 29th June Evening Shift

2.2 g of nitrous oxide (N_2O) gas is cooled at a constant pressure of 1 atm from 310 K to 270 K causing the compression of the gas from 217.1 mL to 167.75 mL. The change in internal energy of the process, ΔU is '$-x$' J. The value of 'x' is _____. [nearest integer]

(Given : atomic mass of N = 14 g mol^{-1} and of O = 16 g mol^{-1}. Molar heat capacity of N_2O is 100 J K^{-1} mol^{-1})

Open solution

Q68. JEE Main 2022 (Online) 29th June Morning Shift

17.0 g of NH_3 completely vapourises at $-33.42°C$ and 1 bar pressure and the enthalpy change in the process is 23.4 kJ mol^{-1}. The enthalpy change for the vapourisation of 85 g of NH_3 under the same conditions is _____ kJ.

Open solution

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Q69. JEE Main 2022 (Online) 28th June Evening Shift

For combustion of one mole of magnesium in an open container at 300 K and 1 bar pressure, $\Delta_C H^\Theta$ = -601.70 kJ mol^{-1}, the magnitude of change in internal energy for the reaction is _____ kJ. (Nearest integer)

(Given : R = 8.3 J K^{-1} mol^{-1})

Open solution

Q70. JEE Main 2022 (Online) 28th June Morning Shift

4.0 L of an ideal gas is allowed to expand isothermally into vacuum until the total volume is 20 L. The amount of heat absorbed in this expansion is _____ L atm.

Open solution

Q71. JEE Main 2022 (Online) 27th June Evening Shift

When 5 moles of He gas expand isothermally and reversibly at 300 K from 10 litre to 20 litre, the magnitude of the maximum work obtained is _____ J. [nearest integer] (Given : R = 8.3 J K^{-1} mol^{-1} and log 2 = 0.3010)

Open solution

Q72. JEE Main 2022 (Online) 26th June Evening Shift

A fish swimming in water body when taken out from the water body is covered with a film of water of weight 36 g. When it is subjected to cooking at 100°C, then the internal energy for vaporization in kJ mol^{-1} is _____. [nearest integer]

[Assume steam to be an ideal gas. Given $\Delta_{vap}H^\Theta$ for water at 373 K and 1 bar is 41.1 kJ mol^{-1} ; R = 8.31 J K^{-1} mol^{-1}]

Open solution

Q73. JEE Main 2022 (Online) 26th June Morning Shift

For complete combustion of methanol

$CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

the amount of heat produced as measured by bomb calorimeter is 726 kJ mol^{-1} at 27°C. The enthalpy of combustion for the reaction is $-x$ kJ mol^{-1}, where x is _____. (Nearest integer)

(Given : R = 8.3 JK^{-1} mol^{-1})

Open solution

Q74. JEE Main 2022 (Online) 25th June Morning Shift

The standard entropy change for the reaction

$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ is -550 J K^{-1} at 298 K.

[Given : The standard enthalpy change for the reaction is -165 kJ mol^{-1}]. The temperature in K at which the reaction attains equilibrium is _____. (Nearest Integer)

Open solution

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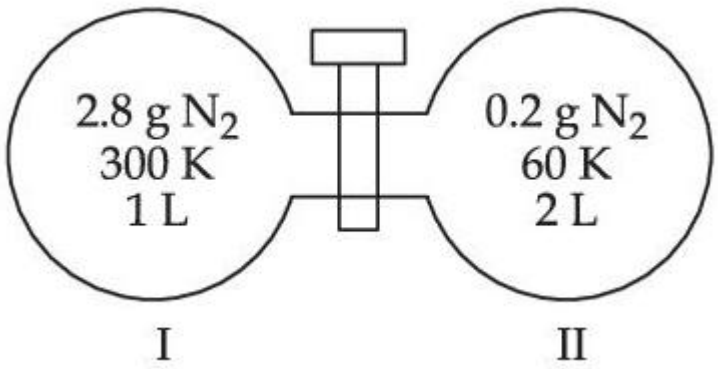
Q75. JEE Main 2021 (Online) 1st September Evening Shift

For the reaction, $2NO_2(g) \rightleftharpoons N_2O_4(g)$, when $\Delta S = -176.0\ JK^{-1}$ and $\Delta H = -57.8\ kJ\ mol^{-1}$, the magnitude of ΔG at 298 K for the reaction is _____ $kJ\ mol^{-1}$. (Nearest integer)

Open solution

Q76. JEE Main 2021 (Online) 27th August Evening Shift

Two flasks I and II shown below are connected by a valve of negligible volume.



When the valve is opened, the final pressure of the system in bar is $x \times 10^{-2}$. The value of x is _____. (Integer answer)

[Assume - Ideal gas; 1 bar = 10^5 Pa; Molar mass of N_2 = 28.0 g mol^{-1}; R = 8.31 J mol^{-1} K^{-1}]

Open solution

Q77. JEE Main 2021 (Online) 27th August Evening Shift

Data given for the following reaction is as follows :

$FeO_{(s)} + C_{(graphite)} \rightarrow Fe_{(s)} + CO_{(g)}$

Substance	ΔH° (kJ mol^{-1})	ΔS° (J mol^{-1} K^{-1})
$FeO_{(s)}$	-266.3	57.49
$C_{(graphite)}$	0	5.74
$Fe_{(s)}$	0	27.28
$CO_{(g)}$	-110.5	197.6

The minimum temperature in K at which the reaction becomes spontaneous is _____. (Integer answer)

Open solution

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Q78. JEE Main 2021 (Online) 27th August Morning Shift

200 mL of 0.2 M HCl is mixed with 300 mL of 0.1 M NaOH. The molar heat of neutralization of this reaction is -57.1 kJ. The increase in temperature in °C of the system on mixing is x $\times$ 10^{-2}. The value of x is _____. (Nearest integer)

[Given : Specific heat of water = 4.18 J g^{-1} K^{-1}, Density of water = 1.00 g cm^{-3}]

[Assume no volume change on mixing]

Open solution

Q79. JEE Main 2021 (Online) 26th August Evening Shift

For water Δ_{vap} H = 41 kJ mol^{-1} at 373 K and 1 bar pressure. Assuming that water vapour is an ideal gas that occupies a much larger volume than liquid water, the internal energy change during evaporation of water is _____ kJ mol^{-1}

[Use : R = 8.3 J mol^{-1} K^{-1}]

Open solution

Q80. JEE Main 2021 (Online) 26th August Morning Shift

The Born-Haber cycle for KCl is evaluated with the following data :

$\Delta_f H^\Theta$ for KCl = -436.7 kJ mol^{-1} ;

$\Delta_{sub} H^\Theta$ for K = 89.2 kJ mol^{-1} ;

$\Delta_{ionization} H^\Theta$ for K = 419.0 kJ mol^{-1} ;

$\Delta_{electron\ gain} H^\Theta$ for $Cl_{(g)}$ = -348.6 kJ mol^{-1} ;

$\Delta_{bond} H^\Theta$ for Cl_2 = 243.0 kJ mol^{-1}

The magnitude of lattice enthalpy of KCl in kJ mol^{-1} is _____ (Nearest integer)

Open solution

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Q81. JEE Main 2021 (Online) 27th July Evening Shift

When 400 mL of 0.2 M H_2SO_4 solution is mixed with 600 mL of 0.1 M NaOH solution, the increase in temperature of the final solution is _____ $\times 10^{-2}$ K. (Round off to the nearest integer).

[Use : H^+ (aq) + OH^- (aq) $\rightarrow$ H_2O : $\Delta_\gamma H = -57.1$ kJ mol^{-1}]

Specific heat of H_2O = 4.18 J K^{-1} g^{-1}

density of H_2O = 1.0 g cm^{-3}

Assume no change in volume of solution on mixing.

Open solution

Q82. JEE Main 2021 (Online) 27th July Morning Shift

For water at 100° C and 1 bar,

$\Delta_{vap} H - \Delta_{vap} U$ = _____ $\times 10^2$ J mol^{-1}. (Round off to the Nearest Integer)

[Use : R = 8.31 J mol^{-1} K^{-1}]

[Assume volume of H_2O(l) is much smaller than volume of H_2O(g). Assume H_2O(g) treated as an ideal gas]

Open solution

Q83. JEE Main 2021 (Online) 25th July Evening Shift

A system does 200 J of work and at the same time absorbs 150 J of heat. The magnitude of the change in internal energy is _____ J. (Nearest integer)

Open solution

Q84. JEE Main 2021 (Online) 25th July Morning Shift

At 298 K, the enthalpy of fusion of a solid (X) is 2.8 kJ mol^{-1} and the enthalpy of vaporisation of the liquid (X) is 98.2 kJ mol^{-1}. The enthalpy of sublimation of the substance (X) in kJ mol^{-1} is _____. (in nearest integer)

Open solution

Q85. JEE Main 2021 (Online) 22th July Evening Shift

If the standard molar enthalpy change for combustion of graphite powder is -2.48×10^2 kJ mol^{-1}, the amount of heat generated on combustion of 1 g of graphite powder is _____ kJ. (Nearest integer)

Open solution

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Q86. JEE Main 2021 (Online) 20th July Evening Shift

For a given chemical reaction A $\rightarrow$ B at 300 K the free energy change is -49.4 kJ mol^{-1} and the enthalpy of reaction is 51.4 kJ mol^{-1}. The entropy change of the reaction is _____ JK^{-1} mol^{-1}.

Open solution

Q87. JEE Main 2021 (Online) 20th July Morning Shift

An average person needs about 10000 kJ energy per day. The amount of glucose (molar mass = 180.0 g mol^{-1}) needed to meet this energy requirement is _____ g.

(Use : $\Delta_C H$(glucose) = -2700 kJ mol^{-1})

Open solution

Q88. JEE Main 2021 (Online) 18th March Morning Shift

For the reaction $C_2H_6 \rightarrow C_2H_4 + H_2$

the reaction enthalpy $\Delta_r H$ = _____ kJ mol^{-1}. (Round off to the Nearest Integer).

[Given : Bond enthalpies in kJ mol^{-1} : C-C : 347, C = C : 611; C-H : 414, H-H : 436]

Open solution

Q89. JEE Main 2021 (Online) 17th March Morning Shift

The standard enthalpies of formation of Al_2O_3 and CaO are -1675 kJ mol^{-1} and -635 kJ mol^{-1} respectively.

For the reaction

$3CaO + 2Al \rightarrow 3Ca + Al_2O_3$ the standard reaction enthalpy $\Delta_r H^0$ = _____ kJ.

(Round off to the Nearest Integer)

Open solution

Q90. JEE Main 2021 (Online) 16th March Evening Shift

At 25°C, 50 g of iron reacts with HCl to form $FeCl_2$. The evolved hydrogen gas expands against a constant pressure of 1 bar. The work done by the gas during this expansion is _____ J. (Round off to the Nearest Integer).

[Given : R = 8.314 J mol^{-1} K^{-1}. Assume, hydrogen is an ideal gas] [Atomic mass of Fe is 55.85 u]

Open solution

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Q91. JEE Main 2021 (Online) 26th February Evening Shift

The average S-F bond energy in kJ mol^{-1} of SF_6 is _____. (Rounded off to the nearest integer)

[Given : The values of standard enthalpy of formation of SF_6(g), S(g) and F(g) are - 1100, 275 and 80 kJ mol^{-1} respectively.]

Open solution

Q92. JEE Main 2021 (Online) 26th February Morning Shift

For a chemical reaction A + B ⇌ C + D

($\Delta_r H^\Theta$ = 80 kJ mol^{-1}) the entropy change $\Delta_r S^\Theta$ depends on the temperature T (in K) as $\Delta_r S^\Theta$ = 2T (J $K^{-1}mol^{-1}$).

Minimum temperature at which it will become spontaneous is _____ K. (Integer)

Open solution

Q93. JEE Main 2021 (Online) 25th February Evening Shift

Five moles of an ideal gas at 293 K is expanded isothermally from an initial pressure of 2.1 MPa to 1.3 MPa against at constant external pressure 4.3 MPa. The heat transferred in this process is _____ kJ mol^{-1}. (Rounded off to the nearest integer) [Use R = 8.314 J $mol^{-1}K^{-1}$]

Open solution

Q94. JEE Main 2021 (Online) 25th February Morning Shift

The reaction of cyanamide, $NH_2CN_{(s)}$ with oxygen was run in a bomb calorimeter and ΔU was found to be -742.24 kJ mol^{-1}. The magnitude of ΔH_{298} for the reaction

$$NH_2CN_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow N_{2(g)} + CO_{2(g)} + H_2O_{(l)}$$

is _____ kJ. (Rounded off to the nearest integer) [Assume ideal gases and R = 8.314 J mol^{-1} K^{-1}]

Open solution

Q95. JEE Main 2021 (Online) 25th February Morning Shift

The ionization enthalpy of Na^+ formation from $Na_{(g)}$ is 495.8 kJ mol^{-1}, while the electron gain enthalpy of Br is -325.0 kJ mol^{-1}. Given the lattice enthalpy of NaBr is -728.4 kJ mol^{-1}. The energy for the formation of NaBr ionic solid is (−) _____ $\times 10^{-1}$ kJ mol^{-1}.

Open solution

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Q96. JEE Main 2021 (Online) 24th February Evening Shift

Assuming ideal behaviour, the magnitude of log K for the following reaction at 25°C is x $\times$ 10^{-1}. The value of x is _____. (Integer answer)

$3HC \equiv CH_{(g)} \rightleftharpoons C_6H_{6(l)}$

[Given : $\Delta_f G^o(HC \equiv CH) = -2.04 \times 10^5$ J mol^{-1} ; $\Delta_f G^o(C_6H_6) = -1.24 \times 10^5$ J mol^{-1} ; R = 8.314 J K^{-1} mol^{-1}]

Open solution

Q97. JEE Main 2020 (Online) 5th September Evening Slot

For a dimerization reaction,

$2A(g) \rightarrow A_2(g)$

at 298 K, ΔU^o = −20 kJ mol^{-1}, ΔS^o = −30 JK^{-1} mol^{-1},

then the ΔG^o will be _____ J.

Open solution

Q98. JEE Main 2020 (Online) 2nd September Evening Slot

The heat of combustion of ethanol into carbon dioxide and water is − 327 kcal at constant pressure. The heat evolved (in cal) at constant volume and 27°C (if all gases behave ideally) is (R = 2 cal mol^{-1} K^{-1}) _____.

Open solution

Q99. JEE Main 2020 (Online) 2nd September Morning Slot

The internal energy change (in J) When 90 g of water undergoes complete evaporation at 100°C is _____.

(Given : ΔH_{vap} for water at 373 K = 41 kJ/mol,

R = 8.314 JK^{-1} mol^{-1})

Open solution

Q100. JEE Main 2020 (Online) 8th January Evening Slot

At constant volume, 4 mol of an ideal gas when heated from 300 K to 500K changes its internal energy by 5000 J. The molar heat capacity at constant volume is _____.

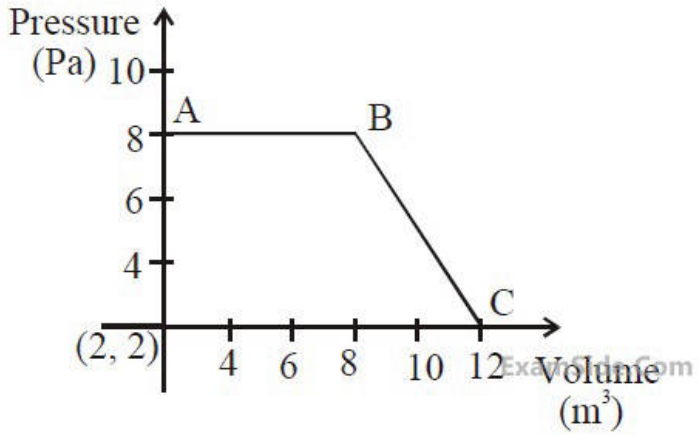
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Q101. JEE Main 2020 (Online) 8th January Morning Slot

The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is _____.

Open solution



Q102. JEE Main 2020 (Online) 7th January Evening Slot

The standard heat of formation $\left(\Delta_f H^0_{298}\right)$ of ethane (in kJ/mol), if the heat of combustion of ethane, hydrogen and graphite are - 1560, -393.5 and -286 kJ/mol, respectively is :

Open solution

Q103. JEE Main 2020 (Online) 7th January Morning Slot

For the reaction :

A(l) $\rightarrow$ 2B(g)

$\Delta U = 2.1\, kcal$, $\Delta S = 20\, cal\, K^{-1}$ at 300 K

Hence ΔG in kcal is :

Open solution

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MCQ (Single Correct Answer) 141 questions

Q104. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

Match List - I with List - II.

Given V_1 and V_2 are initial and final volumes respectively.

List - I (Isothermal process)		List - II (Expression)	
A.	Reversible expansion	I.	$q = 0$
B.	Free expansion	II.	$q = nRT \ln \frac{V_2}{V_1}$
C.	Irreversible Compression	III.	$w = -p_{ext}(V_1 - V_2)$
D.	Cyclic reversible	IV.	$\frac{q_{rev}}{T} = 0$

Choose the correct answer from the options given below :

(A) A-II, B-III, C-I, D-IV

(B) A-II, B-I, C-IV, D-III

(C) A-II, B-I, C-III, D-IV

(D) A-I, B-II, C-III, D-IV

Q105. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

Arrange the following isothermal processes in order of the magnitude of the work $(p - V)$ involved between states 1 and 2.

A. Expansion in single stage w_A

B. Expansion in multi stages w_B

C. Compression in single stage w_C

D. Compression in multi stages w_D

Choose the correct option.

(A) $|w_B| > |w_A| > |w_C| > |w_D|$

(B) $|w_C| > |w_D| > |w_A| > |w_B|$

(C) $|w_C| > |w_D| > |w_B| > |w_A|$

(D) $|w_B| > |w_A| > |w_D| > |w_C|$

Q106. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Consider the following data for the reaction

$X_2(g) + Y_2(g) \rightleftharpoons 2XY(g)$

at $600\ K^\circ$. The $\Delta_r G^\ominus$ (in $kJmol^{-1}$) for the reaction is :

Compound	$\Delta_f H^\ominus_{600\ K}$ $(kJ\ mol^{-1})$	$S^\ominus_{600\ K}$ $(J\ mol^{-1}\ K^{-1})$
$XY(g)$	42	200
$X_2(g)$	8	140
$Y_2(g)$	80	250

(A) -21000 (B) -10 (C) -1000 (D) -9.012

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Q107. JEE Main 2026 (Online) 5th April Evening Shift

The correct order of molar heat capacities measured at 298 K and 1 bar is :

(A) Copper(s) $>$ Bromine(l) $>$ Helium(g)

(B) Bromine(l) > Copper(s) > Helium(g)

(C) Helium(g) > Bromine(l) > Copper(s)

(D) Helium(g) > Bromine(l) = Copper(s)

Open solution

Q108. JEE Main 2026 (Online) 4th April Morning Shift

Given below are two statements :

Statement I : For an ideal gas, heat capacity at constant volume is always greater than the heat capacity at constant pressure.

Statement II : In a constant volume process, no work is produced and all the heat withdrawn goes into the chaotic motion and is reflected by a temperature increase of the ideal gas.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Open solution

Q109. JEE Main 2026 (Online) 2nd April Evening Shift

Gas 'A' undergoes change from state 'X' to state 'Y'. In this process, the heat absorbed and work done by the gas is 10 J and 18 J respectively. Now gas is brought back to state 'X' by another process during which 6 J of heat is evolved. In the reverse process of 'Y' to 'X',

(A) 18 J of the work is done by the gas 'A'.

(B) 2 J of the work is done by the gas 'A'.

(C) 12 J of the work is done on the gas 'A' by the surrounding.

(D) 14 J of the work is done on the gas 'A' by the surrounding.

Open solution

Q110. JEE Main 2026 (Online) 2nd April Morning Shift

Consider the following data.

(i) $2Al(s) + 6HCl(aq) \rightarrow Al_2Cl_6(aq) + 3H_2(g) + 1200$ kJ/mol

(ii) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g) + 164$ kJ/mol

(iii) $HCl(g) + aq \rightarrow HCl(aq) + 83$ kJ/mol

(iv) $Al_2Cl_6(s) + aq \rightarrow Al_2Cl_6(aq) + 663$ kJ/mol

The enthalpy of formation of anhydrous solid Al_2Cl_6 is :

(A) -648 kJ mol^{-1} (B) -1350 kJ mol^{-1} (C) -2002 kJ mol^{-1} (D) -1527 kJ mol^{-1}

Open solution

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Q111. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

The plot of $\log_{10} K$ vs $\frac{1}{T}$ gives a straight line. The intercept and slope respectively are (where K is equilibrium constant).

(A) $-\frac{\Delta S^\circ R}{2.303}, \frac{\Delta H^\circ R}{2.303}$

(B) $-\frac{\Delta H^\circ}{2.303R}, \frac{\Delta S^\circ}{2.303R}$

(C) $\frac{\Delta S^\circ}{2.303R}, -\frac{\Delta H^\circ}{2.303R}$

(D) $\frac{2.303R}{\Delta H^\circ}, \frac{2.303R}{\Delta S^\circ}$

Q112. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

$20.0\mathrm{dm}^3$ of an ideal gas ' X ' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until pressure of the gas is 0.2 MPa . Which of the following option is correct?

(Given: $\log 2 = 0.3010$ and $\log 5 = 0.6989$)

(A) $\mathrm{w} = -9.1\ \mathrm{kJ}, \Delta\mathrm{U} = 0, \Delta\mathrm{H} = 0, \mathrm{q} = 9.1\ \mathrm{kJ}$

(B) $\mathrm{w} = 9.1\ \mathrm{J}, \Delta\mathrm{U} = 9.1\ \mathrm{J}, \Delta\mathrm{H} = 0; \mathrm{q} = 0$

(C) $\mathrm{w} = -3.9\ \mathrm{kJ}, \Delta\mathrm{U} = 0, \Delta\mathrm{H} = 0; \mathrm{q} = 3.9\ \mathrm{kJ}$

(D) $\mathrm{w} = +4.1\ \mathrm{kJ}, \Delta\mathrm{U} = 0, \Delta\mathrm{H} = 0; \mathrm{q} = -4.1\ \mathrm{kJ}$

Q113. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

The heat of atomisation of methane and ethane are ' x ' kJmol^{-1} and ' y ' kJmol^{-1} respectively. The longest wavelength (λ) of light capable of breaking the $\mathrm{C-C}$ bond can be expressed in SI unit as :

(A) $\frac{\mathrm{N_A hc}}{250(y-6x)}$

(B) $\mathrm{N_A hc}\left(y-\frac{6x}{4}\right)^{-1}$

(C) $\frac{\mathrm{hc}}{1000}\left(\frac{y-6x}{4}\right)^{-1}$

(D) $\frac{\mathrm{N_A hc}}{250(4y-6x)}$

Q114. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Match the LIST-I with LIST-II

List-I Isothermal process for ideal gas system		List-II Work done ($V_f > V_i$)	
A.	Reversible expansion	I.	$w = 0$
B.	Free expansion	II.	$w = -nRT\ln\left(\frac{V_f}{V_i}\right)$
C.	Irreversible expansion	III.	$w = -p_{ex}(V_f - V_i)$
D.	Irreversible compression	IV.	$w = -p_{ex}(V_i - V_f)$

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II

(B) A-I, B-III, C-II, D-IV

(C) A-II, B-I, C-III, D-IV

(D) A-IV, B-II, C-III, D-I

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Q115. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

A cup of water at $5^{\circ}\mathrm{C}$ (system) is placed in a microwave oven and the oven is turned on for one minute during which the water begins to boil. Which of the following options is true?

(A) $q = +ve, w = -ve, \Delta U = +ve$

(B) $q = +ve, w = 0, \Delta U = -ve$

(C) $q = +ve, w = -ve, \Delta U = -ve$

(D) $q = -ve, w = -ve, \Delta U = -ve$

Q116. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Match the LIST-I with LIST-II

List-I **Thermodynamic Process**	**List-II** **Magnitude in kJ**
A. Work done in reversible, isothermal expansion of 2 mol of ideal gas from $2\,\mathrm{dm}^3$ to $20\,\mathrm{dm}^3$ at $300\,\mathrm{K}$.	I. 4
B. Work done in irreversible isothermal expansion of 1 mol ideal gas from $1\,\mathrm{m}^3$ to $3\,\mathrm{m}^3$ at $300\,\mathrm{K}$ against a constant pressure of $3\,\mathrm{kPa}$.	II. 11.5
C. Change in internal energy for adiabatic expansion of a 1 mol ideal gas with change of temperature $= 320\,\mathrm{K}$ and $\overline{C}_V = \frac{3}{2}R$.	III. 6
D. Change in enthalpy at constant pressure of 1 mol ideal gas with change of temperature $= 337\,\mathrm{K}$ and $\overline{C}_p = \frac{5}{2}R$.	IV. 7

(A) A-I, B-II, C-III, D-IV

(B) A-III, B-II, C-IV, D-I

(C) A-II, B-I, C-III, D-IV

(D) A-II, B-III, C-I, D-IV

Q117. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

Consider the following data :

$\Delta_f H^{\circ}$ (methane, g) = $-X$ kJ mol^{-1}

Enthalpy of sublimation of graphite = Y kJ mol^{-1}

Dissociation enthalpy of $H_2 = Z$ kJ mol^{-1}

The bond enthalpy of C–H bond is given by :

(A) $\frac{X + Y + 2Z}{4}$

(B) $\frac{X + Y + 4Z}{2}$

(C) $\frac{-X + Y + Z}{4}$

(D) $X + Y + Z$

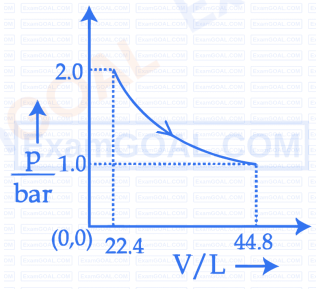
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Q118. JEE Main 2026 (Online) 21st January Morning Shift

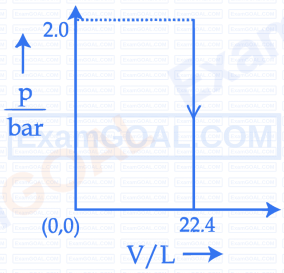
Which of the following graphs between pressure ' p ' versus volume ' V ' represents the maximum work done?

Open solution

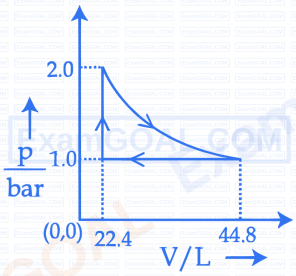
(A)



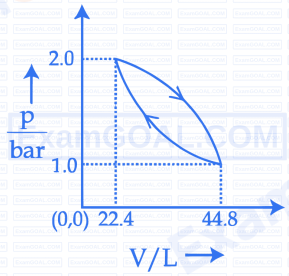
(B)



(C)



(D)



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Q119. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

For the reaction, $N_2O_4 \rightleftharpoons 2NO_2$, graph is plotted as shown below. Identify correct statements.

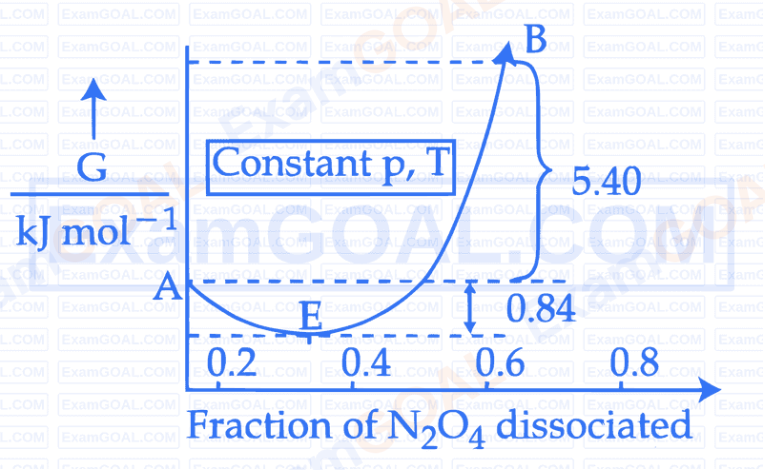
A. Standard free energy change for the reaction is $-5.40\ kJ\ mol^{-1}$.

B. As $\Delta G^{\ominus}$ in graph is positive, N_2O_4 will not dissociate into NO_2 at all.

C. Reverse reaction will go to completion.

D. When 1 mole of N_2O_4 changes into equilibrium mixture, value of $\Delta G^{\ominus} = -0.84\ kJ\ mol^{-1}$

E. When 2 mole of NO_2 changes into equilibrium mixture, $\Delta G^{\ominus}$ for equilibrium mixture is $-6.24\ kJ\ mol^{-1}$.



Choose the correct answer from the options given below :

(A) C and E only (B) D and E only (C) A and D only (D) B and C only

Q120. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

The hydration energies of K^+ and Cl^- are $-x$ and $-y$ kJ/mol respectively. If the lattice energy of KCl is $-z$ kJ/mol, then the heat of solution of KCl is :

(A) $x + y + z$ (B) $z - (x + y)$ (C) $-z - (x + y)$ (D) $x - y - z$

Q121. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

The correct statement amongst the following is :

(A) The standard state of a pure gas is the pure gas at a pressure of 1 bar and temperature 273 K.

(B) $\Delta_f H^{\circ}_{500}$ is zero for $O_2(g)$.

(C) $\Delta_f H^{\circ}_{298}$ is zero for $O(g)$.

(D) The term 'standard state' implies that the temperature is 0°C.

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Q122. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Total enthalpy change for freezing of 1 mol of water at $10^{\circ}\mathrm{C}$ to ice at $-10^{\circ}\mathrm{C}$ is _____

(Given: $\Delta_{\text{fus}}\,\mathrm{H} = x\,\mathrm{kJ/mol}$

$\mathrm{C_p\,[H_2O(l)]} = y\,\mathrm{J\,mol^{-1}\,K^{-1}}$

$\mathrm{C_p\,[H_2O(\,s)]} = z\,\mathrm{J\,mol^{-1}\,K^{-1}}$

(A) $-x-10y-10z$ (B) $x-10y-10z$ (C) $-10(100x+y+z)$ (D) $10(100\mathrm{x}+y+z)$

Q123. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Consider the given data :

(a) $\mathrm{HCl(g)} + 10\mathrm{H_2O(l)} \rightarrow \mathrm{HCl.10H_2O}\,\Delta\mathrm{H} = -69.01\,\mathrm{kJ\,mol^{-1}}$

(b) $\mathrm{HCl(g)} + 40\mathrm{H_2O(l)} \rightarrow \mathrm{HCl.40H_2O}\,\Delta\mathrm{H} = -72.79\,\mathrm{kJ\,mol^{-1}}$

Choose the correct statement :

(A) The heat of dilution for the HCl ($\mathrm{HCl.10H_2O}$ to $\mathrm{HCl.40H_2O}$) is $3.78\,\mathrm{kJ\,mol^{-1}}$.

(B) Dissolution of gas in water is an endothermic process.

(C) The heat of solution depends on the amount of solvent.

(D) The heat of formation of HCl solution is represented by both (a) and (b).

Q124. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

One mole of an ideal gas expands isothermally and reversibly from $10\mathrm{dm^3}$ to $20\mathrm{dm^3}$ at 300 K . $\Delta\mathrm{U}, \mathrm{q}$ and work done in the process respectively are

Given: $\mathrm{R} = 8.3\,\mathrm{J\,K^{-1}\,mol^{-1}}$

$\ln 10 = 2.3$

$\log 2 = 0.30$

$\log 3 = 0.48$

(A) $0, 21.84\,\mathrm{kJ}, -1.726\,\mathrm{J}$ (B) $0, 21.84\,\mathrm{kJ}, 21.84\,\mathrm{kJ}$

(C) $0, 1.718\,\mathrm{kJ}, -1.718\,\mathrm{kJ}$ (D) $0, -17.18\,\mathrm{kJ}, 1.718\,\mathrm{J}$

Q125. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Let us consider a reversible reaction at temperature, T. In this reaction, both $\Delta\mathrm{H}$ and $\Delta\mathrm{S}$ were observed to have positive values. If the equilibrium temperature is Te , then the reaction becomes spontaneous at:

(A) $\mathrm{Te} > \mathrm{T}$ (B) $\mathrm{T} > \mathrm{Te}$ (C) $\mathrm{T} = \mathrm{Te}$ (D) $\mathrm{Te} = 5\,\mathrm{T}$

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Q126. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Given below are two statements:

Statement I : When a system containing ice in equilibrium with water (liquid) is heated, heat is absorbed by the system and there is no change in the temperature of the system until whole ice gets melted.

Statement II : At melting point of ice, there is absorption of heat in order to overcome intermolecular forces of attraction within the molecules of water in ice and kinetic energy of molecules is not increased at melting point.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

Q127. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Arrange the following in order of magnitude of work done by the system/on the system at constant temperature.

(a) $|w_{\text{reversible}}|$ for expansion in infinite stages.

(b) $|w_{\text{irreversible}}|$ for expansion in single stage.

(c) $|\mathrm{w}_{\text{reversible}}|$ for compression in infinite stages.

(d) $|w_{\text{irreversible}}|$ for compression in single stage.

Choose the correct answer from the options given below :

(A) $\mathrm{d} > \mathrm{c} = \mathrm{a} > \mathrm{b}$

(B) $\mathrm{c} = \mathrm{a} > \mathrm{d} > \mathrm{b}$

(C) $a > c > b > d$

(D) $a > b > c > d$

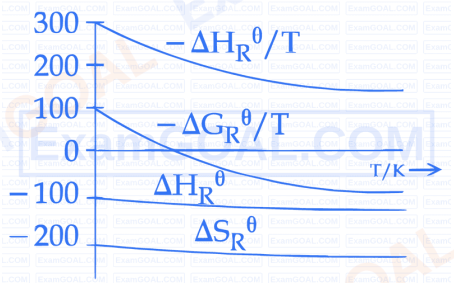
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Q128. JEE Main 2025 (Online) 2nd April Evening Shift

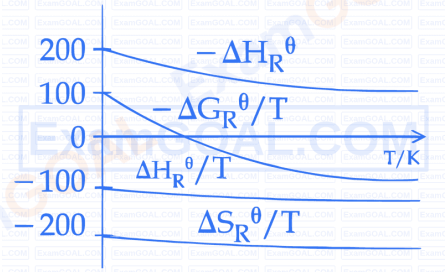
Which of the following graphs correctly represents the variation of thermodynamic properties of Haber's process?

Open solution

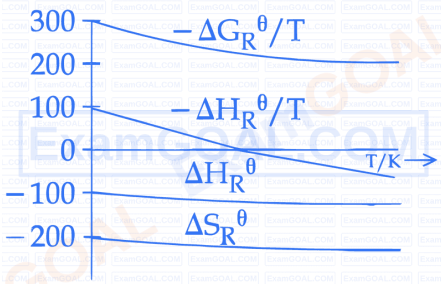
(A)



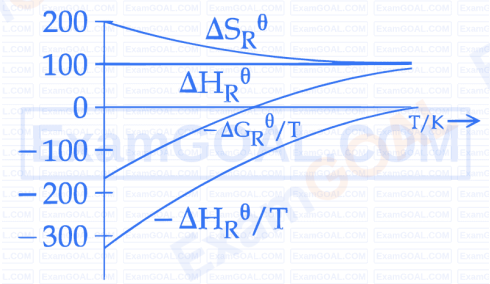
(B)



(C)

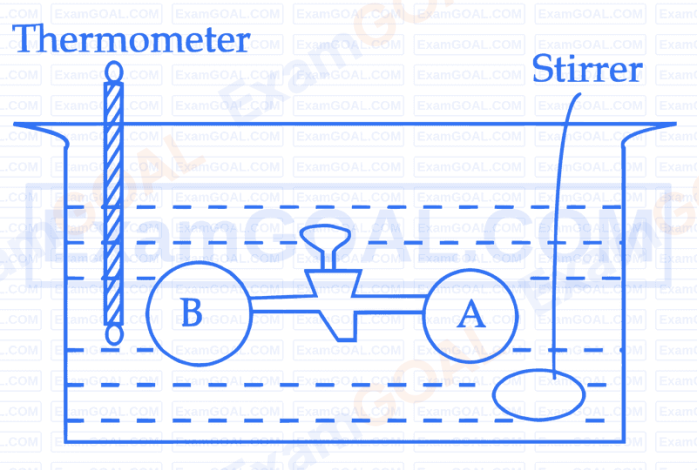


(D)



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Q129. JEE Main 2025 (Online) 2nd April Morning Shift



Open solution

Two vessels A and B are connected via stopcock. The vessel A is filled with a gas at a certain pressure. The entire assembly is immersed in water and is allowed to come to thermal equilibrium with water. After opening the stopcock the gas from vessel A expands into vessel B and no change in temperature is observed in the thermometer. Which of the following statement is true ?

(A) $\mathrm{dq} \neq 0$

(B) The pressure in the vessel B before opening the stopcock is zero

(C) $\mathrm{dw} \neq 0$

(D) $\mathrm{dU} \neq 0$

Q130. JEE Main 2025 (Online) 29th January Evening Shift

If C (diamond) $\rightarrow$ C (graphite) $+X\ \mathrm{kJ\ mol^{-1}}$

C (diamond) $+\mathrm{O_2(g)} \rightarrow \mathrm{CO_2(g)} + \mathrm{YkJmol^{-1}}$

C (graphite) $+\mathrm{O_2(g)} \rightarrow \mathrm{CO_2(g)} + \mathrm{ZkJmol^{-1}}$

at constant temperature. Then

Open solution

(A) −X = Y + Z (B) X = Y − Z (C) X = −Y + Z (D) X = Y + Z

Q131. JEE Main 2025 (Online) 29th January Morning Shift

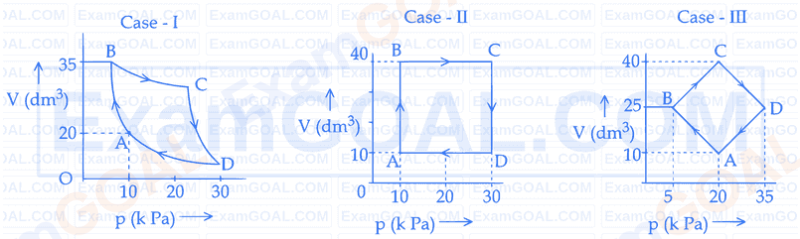
500 J of energy is transferred as heat to 0.5 mol of Argon gas at 298 K and 1.00 atm. The final temperature and the change in internal energy respectively are: Given: R = 8.3 J K^{-1} mol^{-1}

Open solution

(A) 368 K and 500 J (B) 348 K and 300 J (C) 378 K and 300 J (D) 378 K and 500 J

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Q132. JEE Main 2025 (Online) 28th January Evening Shift



Open solution

An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A → B → C → D → A as shown in the three cases above.

Choose the **correct** option regarding ΔU :

(A) $\Delta U(\text{ Case-III }) > \Delta U(\text{ Case-II }) > \Delta U(\text{ Case-I })$

(B) $\Delta U(\text{ Case-I }) > \Delta U(\text{ Case-II }) > \Delta U(\text{ Case-III })$

(C) $\Delta U(\text{ Case-I }) = \Delta U(\text{ Case-II }) = \Delta U(\text{ Case-III })$

(D) $\Delta U(\text{ Case-I }) > \Delta U(\text{ Case-III }) > \Delta U(\text{ Case-II })$

Q133. JEE Main 2025 (Online) 28th January Morning Shift

Ice and water are placed in a closed container at a pressure of 1 atm and temperature 273.15 K . If pressure of the system is increased 2 times, keeping temperature constant, then identify correct observation from following

Open solution

(A) Liquid phase disappears completely.

(B) The amount of ice decreases.

(C) The solid phase (ice) disappears completely.

(D) Volume of system increases .

Q134. JEE Main 2025 (Online) 24th January Evening Shift

$$\mathrm{S}(\mathrm{g}) + \frac{3}{2}\mathrm{O}_2(\mathrm{g}) \rightarrow \mathrm{SO}_3(\mathrm{g}) + 2x\,\mathrm{kcal}$$

$$\mathrm{SO}_2(\mathrm{g}) + \frac{1}{2}\mathrm{O}_2(\mathrm{g}) \rightarrow \mathrm{SO}_3(\mathrm{g}) + y\,\mathrm{kcal}$$

The heat of formation of $\mathrm{SO}_2(\mathrm{g})$ is given by :

Open solution

(A) $2x + y$ kcal **(B)** $\frac{2x}{y}$ kcal **(C)** $y - 2x$ kcal **(D)** $x + y$ kcal

Q135. JEE Main 2025 (Online) 24th January Evening Shift

Which of the following mixing of 1 M base and 1 M acid leads to the largest increase in temperature?

Open solution

(A) 50 mL HCl and 20 mL NaOH

(B) 30 mL HCl and 30 mL NaOH

(C) 45 mL $\mathrm{CH_3COOH}$ and 25 mL NaOH

(D) 30 mL $\mathrm{CH_3COOH}$ and 30 mL NaOH

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Q136. JEE Main 2025 (Online) 24th January Morning Shift

Let us consider an endothermic reaction which is non-spontaneous at the freezing point of water. However, the reaction is spontaneous at boiling point of water. Choose the correct option.

Open solution

(A) Both ΔH and ΔS are (-ve)

(B) ΔH is $(+ve)$ but ΔS is (-ve)

(C) ΔH is $(-ve)$ but ΔS is (+ve)

(D) Both ΔH and ΔS are (+ve)

Q137. JEE Main 2025 (Online) 23rd January Evening Shift

The effect of temperature on spontaneity of reactions are represented as :

Open solution

	ΔH	ΔS	Temperature	Spontaneity
(A)	+	−	any T	Non spontaneous
(B)	+	+	low T	spontaneous
(C)	−	−	low T	Non spontaneous
(D)	−	+	any T	spontaneous

The incorrect combinations are :

(A) (A) and (C) only **(B)** (B) and (D) only **(C)** (A) and (D) only **(D)** (B) and (C) only

Q138. JEE Main 2025 (Online) 23rd January Morning Shift

Ice at $-5^{\circ}C$ is heated to become vapor with temperature of $110^{\circ}C$ at atmospheric pressure. The entropy change associated with this process can be obtained from

Open solution

(A) $\int_{268\,K}^{273\,K} C_{p,m}dT + \frac{\Delta H_{m,\,fusion}}{T_f} + \frac{\Delta H_{m,\,vaporisation}}{T_b} + \int_{273\,K}^{373\,K} C_{p,m}dT + \int_{373\,K}^{383\,K} C_{p,m}dT$

(B) $\int_{268\,K}^{383\,K} C_p dT + \frac{\Delta H_{melting}}{273} + \frac{\Delta H_{boiling}}{373}$

(C) $\int_{268\,K}^{383\,K} C_p dT + \frac{q_{rev}}{T}$

(D) $\int_{268\,K}^{273\,K} \frac{C_{p,m}}{T}dT + \frac{\Delta H_{m,\,fusion}}{T_f} + \frac{\Delta H_{m,\,vaporisation}}{T_b} + \int_{273\,K}^{373\,K} \frac{C_{p,m}dT}{T} + \int_{373\,K}^{383\,K} \frac{C_{p,m}dT}{T}$

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Q139. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Match List - I with List - II.

	List - I (Partial Derivatives)		List - II (Thermodynamic Quantity)
(A)	$\left(\frac{\partial G}{\partial T}\right)_P$	(I)	Cp
(B)	$\left(\frac{\partial H}{\partial T}\right)_P$	(II)	$-S$
(C)	$\left(\frac{\partial G}{\partial P}\right)_T$	(III)	Cv
(D)	$\left(\frac{\partial U}{\partial T}\right)_V$	(IV)	V

Choose the correct answer from the options given below :

(A) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(B) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(C) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(D) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

Q140. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

A liquid when kept inside a thermally insulated closed vessel at $25^{\circ}C$ was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters ?

(A) $\Delta U = 0, q < 0, w > 0$

(B) $\Delta U > 0, q = 0, w > 0$

(C) $\Delta U = 0, q = 0, w = 0$

(D) $\Delta U < 0, q = 0, w > 0$

Q141. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Enthalpy of neutralisation of strong monobasic acid with strong monoacidic base is always $-57\ kJ\ mol^{-1}$

Reason (R) : Enthalpy of neutralisation is the amount of heat liberated when one mole of H^+ ions furnished by acid combine with one mole of ^-OH ions furnished by base to form one mole of water.

In the light of the above statements, choose the correct answer from the options given below.

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(D) (A) is false but (R) is true

Q142. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following :

(A) $q = 0, \Delta T = 0, w = 0$

(B) $q = 0, \Delta T \neq 0, w = 0$

(C) $q \neq 0, \Delta T = 0, w = 0$

(D) $q = 0, \Delta T < 0, w \neq 0$

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Q143. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Which of the following is not correct?

(A) ΔG is positive for a spontaneous reaction

(B) ΔG is positive for a non-spontaneous reaction

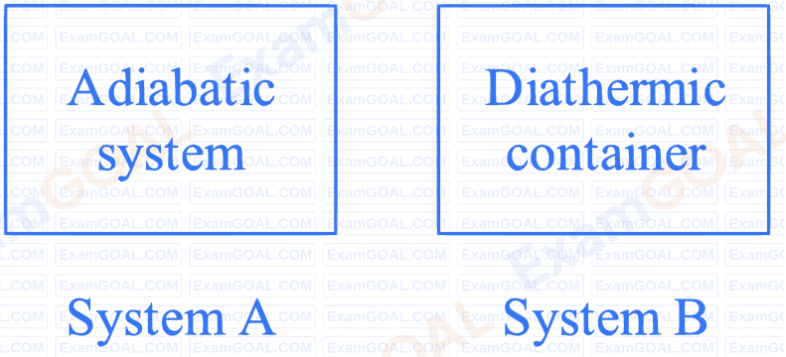
(C) ΔG is zero for a reversible reaction

(D) ΔG is negative for a spontaneous reaction

Q144. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

What happens when methane undergoes combustion in systems A and B respectively?



(A)

System A	System B
Temperature falls	Temperature rises

(B)

System A	System B
Temperature rises	Temperature remains same

(C)

System A	System B
Temperature falls	Temperature remains same

(D)

System A	System B
Temperature remains same	Temperature rises

Q145. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

Given

(A)	$2CO(g) + O_2(g) \rightarrow 2CO_2(g)$		$\Delta H_1^0 = -x \text{ kJ mol}^{-1}$
(B)	$C(\text{graphite}) + O_2(g) \rightarrow CO_2(g)$		$\Delta H_2^0 = -y \text{ kJ mol}^{-1}$

ΔH^0 for the reaction

$C(\text{graphite}) + \frac{1}{2}O_2(g) \rightarrow CO(g)$ is :

(A) $\frac{x-2y}{2}$ (B) $\frac{x+2y}{2}$ (C) $2y - x$ (D) $\frac{2x-y}{2}$

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Q146. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Which of the following relations are correct?

(A) $\Delta U = q + p\Delta V$

(B) $\Delta G = \Delta H - T\Delta S$

(C) $\Delta S = \frac{q_{rev}}{T}$

(D) $\Delta H = \Delta U - \Delta nRT$

Choose the most appropriate answer from the options given below :

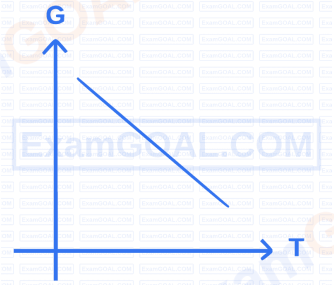
(A) A and B only **(B)** B and C only **(C)** C and D only **(D)** B and D only

Q147. JEE Main 2022 (Online) 30th July Evening Shift

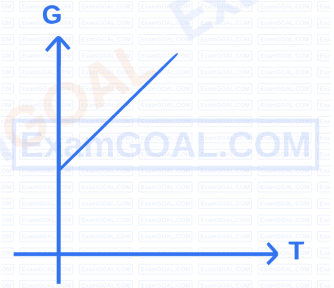
Open solution

Which of the following is the plot of Gibb's free energy (G) vs temperature (T), at constant pressure, for a pure substance?

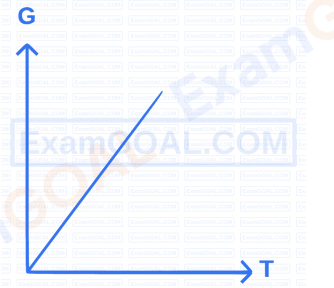
(A)



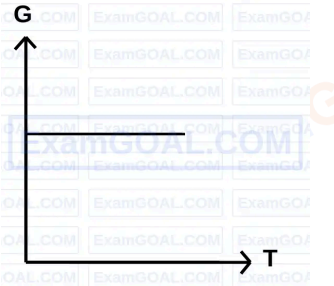
(B)



(C)



(D)



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Q148. JEE Main 2022 (Online) 28th July Evening Shift

Open solution

Given below are two statements: One is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : The reduction of a metal oxide is easier if the metal formed is in liquid state than solid state.

Reason $\mathbf{R}$: The value of ΔG^{Θ} becomes more on negative side as entropy is higher in liquid state than solid state.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct and R is the correct explanation of A

(B) Both A and R are correct but R is NOT the correct explanation of A

(C) A is correct but R is not correct

(D) A is not correct but R is correct

Q149. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

Which of the following relation is not correct?

(A) $\Delta \mathrm{H}=\Delta \mathrm{U}-\mathrm{P} \Delta \mathrm{V}$

(B) $\Delta \mathrm{U}=\mathrm{q}+\mathrm{W}$

(C) $\Delta \mathrm{S}_{\text{sys}}+\Delta \mathrm{S}_{\text{surr}} \geqslant 0$

(D) $\Delta \mathrm{G}=\Delta \mathrm{H}-\mathrm{T} \Delta \mathrm{S}$

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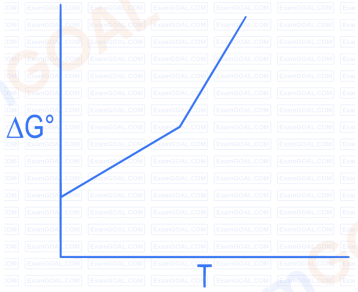
Q150. JEE Main 2022 (Online) 30th June Morning Shift

Open solution

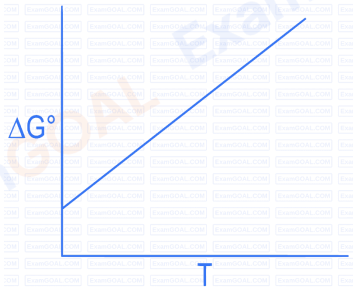
ΔG° vs T plot for the formation of MgO, involving reaction

$2Mg + O_2 \rightarrow 2MgO$, will look like :

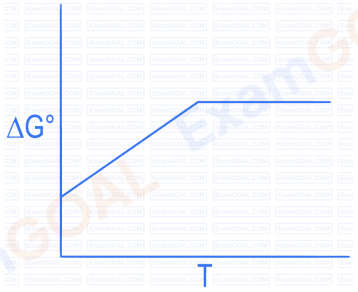
(A)



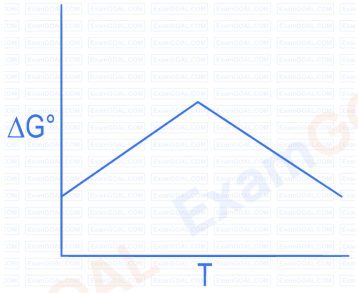
(B)



(C)



(D)



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Q151. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

Match List-I with List-II.

	List - I		List - II
(A)	Spontaneous process	(I)	$\Delta H < 0$
(B)	Process with $\Delta P = 0$, $\Delta T = 0$	(II)	$\Delta G_{T,P} < 0$
(C)	$\Delta H_{reaction}$	(III)	Isothermal and isobaric process
(D)	Exothermic Process	(IV)	[Bond energies of molecules in reactants] − [Bond energies of product molecules

Choose the correct answer from the options given below :

(A) (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

(B) (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

(C) (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

(D) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

Q152. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

At 25°C and 1 atm pressure, the enthalpy of combustion of benzene (l) and acetylene (g) are -3268 kJ mol^{-1} and -1300 kJ mol^{-1}, respectively. The change in enthalpy for the reaction 3 $C_2H_2(g) \rightarrow C_6H_6$ (l), is :

(A) +324 kJ mol^{-1} **(B)** +632 kJ mol^{-1} **(C)** $-$ 632 kJ mol^{-1} **(D)** $-$ 732 kJ mol^{-1}

Q153. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

At 25°C and 1 atm pressure, the enthalpies of combustion are as given below :

Substance	H_2	C (graphite)	$C_2H_6(g)$
$\frac{\Delta_c H^\Theta}{kJ\ mol^{-1}}$	-286.0	-394.0	-1560.0

The enthalpy of formation of ethane is

(A) +54.0 kJ mol^{-1} **(B)** $-$68.0 kJ mol^{-1} **(C)** $-$86.0 kJ mol^{-1} **(D)** +97.0 kJ mol^{-1}

Q154. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

The incorrect expression among the following is :

(A) $\frac{\Delta G_{System}}{\Delta S_{Total}} = -T$ (at constant P)

(B) $K = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$

(C) $K = e^{-\Delta G^\circ / RT}$

(D) For isothermal process $w_{reversible} = -nRT \ln \frac{V_f}{V_i}$

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Q155. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

During which of the following processes, does entropy decrease?

(A) Freezing of water to ice at 0°C

(B) Freezing of water to ice at $-10^{\circ}C$

(C) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

(D) Adsorption of CO(g) on lead surface.

(E) Dissolution of NaCl in water

Choose the correct answer from the options given below :

(A) (A), (C) and (E) only **(B)** (B) and (C) only

(C) (A), (B), (C) and (D) only **(D)** (A) and (E) only

Q156. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

Lattice enthalpy and enthalpy of solution of NaCl are 788 kJ mol^{-1}, and 4 kJ mol^{-1}, respectively. The hydration enthalpy of NaCl is :

(A) -780 kJ mol^{-1} **(B)** -784 kJ mol^{-1} **(C)** 780 kJ mol^{-1} **(D)** 784 kJ mol^{-1}

Q157. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

Five moles of an ideal gas at 1 bar and 298 K is expanded into vacuum to double the volume. The work done is :

(A) Zero **(B)** -RT $\ln \frac{V_2}{V_1}$

(C) $C_V (T_2 - T_1)$ **(D)** $-$ RT $(V_2 - V_1)$

Q158. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

For one mole of an ideal gas, which of these statements must be true?
(a) U and H each depends only on temperature
(b) Compressibility factor z is not equal to 1
(c) $C_{P,m} - C_{V,m} = R$
(d) $dU = C_V dT$ for any process

(A) (a), (c) and (d) **(B)** (a) and (c) **(C)** (c) and (d) **(D)** (b), (c) and (d)

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Q159. JEE Main 2020 (Online) 9th January Evening Slot

The true statement amongst the following is :

(A) S is a function of temperature but ΔS is not a function of temperature.

(B) Both S and ΔS are not functions of temperature.

(C) Both ΔS and S are functions of temperature.

(D) S is not a function of temperature but ΔS is a function of temperature.

Open solution

Q160. JEE Main 2020 (Online) 9th January Morning Slot

If enthalpy of atomisation for $Br_2(l)$ is x kJ/mol and bond enthalpy for Br_2 is y kJ/mol, the relation between them :

(A) does not exist (B) is x < y (C) is x > y (D) is x = y

Open solution

Q161. JEE Main 2019 (Online) 12th April Morning Slot

Enthalpy of sublimation of iodine is 24 cal g^{-1} at 200 ^oC. If specific heat of $I_2(s)$ and I_2 (vap) are 0.055 and 0.031 cal $g^{-1}K^{-1}$ respectively, then enthalpy of sublimation of iodine at 250 ^oC in cal g^{-1} is :

(A) 2.85 (B) 22.8 (C) 11.4 (D) 5.7

Open solution

Q162. JEE Main 2019 (Online) 12th April Morning Slot

An ideal gas is allowed to expand from 1 L to 10 L against a constant external pressure of 1 bar. The work done in kJ is :

(A) +10.0 (B) −0.9 (C) − 2.0 (D) − 9.0

Open solution

Q163. JEE Main 2019 (Online) 10th April Evening Slot

The difference between ΔH and ΔU ($\Delta H - \Delta U$), when the combustion of one mole of heptane(l) is carried out at a temperature T, is equal to :

(A) − 4 RT (B) 3 RT (C) − 3 RT (D) 4 RT

Open solution

Q164. JEE Main 2019 (Online) 10th April Morning Slot

A process will be spontaneous at all temperatures if :

(A) $\Delta H < 0$ and $\Delta S > 0$ (B) $\Delta H < 0$ and $\Delta S < 0$

(C) $\Delta H > 0$ and $\Delta S < 0$ (D) $\Delta H > 0$ and $\Delta S > 0$

Open solution

Q165. JEE Main 2019 (Online) 9th April Evening Slot

During compression of a spring the work done is 10kJ and 2kJ escaped to the surroundings as heat. The change in internal energy, ΔU(inkJ) is :

(A) - 12 (B) 8 (C) - 8 (D) 12

Open solution

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Q166. JEE Main 2019 (Online) 9th April Morning Slot

Among the following, the set of parameters that represents path function, is :
(A) q + w
(B) q
(C) w
(D) H−TS

(A) (B) and (C) **(B)** (A) and (D) **(C)** (B), (C) and (D) **(D)** (A), (B) and (C)

Open solution

Q167. JEE Main 2019 (Online) 8th April Evening Slot

5 moles of an ideal gas at 100 K are allowed to undergo reversible compression till its temperature becomes 200 K. If C_V = 28 $JK^{-1}mol^{-1}$, calculate ΔU and ΔpV for this process. (R = 8.0 JK^{-1} mol^{-1}]

(A) ΔU = 14 kJ; Δ(pV) = 4 kJ
(B) ΔU = 2.8 kJ; Δ(pV) = 0.8 kJ
(C) ΔU = 14 kJ; Δ(pV) = 18 kJ
(D) ΔU = 14 kJ; Δ(pV) = 0.8 kJ

Open solution

Q168. JEE Main 2019 (Online) 8th April Morning Slot

For silver, C_p(J K^{-1} mol^{-1}) = 23 +0.01 T. If the temperature (T) of 3 moles of silver is raised from 300 K to 1000 K at 1 atm pressure, the value of ΔH will be close to :

(A) 62 kJ **(B)** 16 kJ **(C)** 13 kJ **(D)** 21 kJ

Open solution

Q169. JEE Main 2019 (Online) 8th April Morning Slot

Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas? (Assume non-expansion work is zero)

(A) Adiabatic process : $\Delta U = -w$
(B) Cyclic process : $q = -w$
(C) Isochoric process : $\Delta U = q$
(D) Isothermal process : $q = -w$

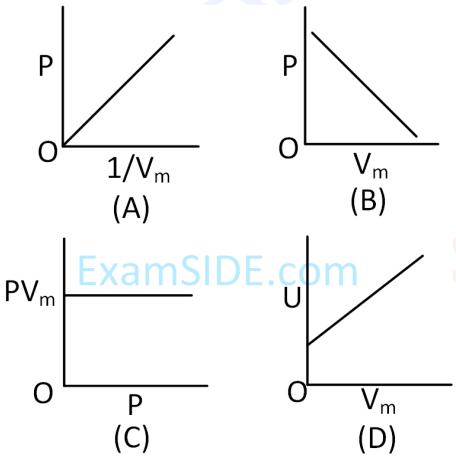
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Q170. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

The combination of plots which does not represent isothermal expansion of an ideal gas is –



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(A) A and D **(B)** B and D **(C)** B and C **(D)** A and C

Q171. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

Given

(i) C (graphite) + $O_2(g) \rightarrow CO_2(g)$; ΔrH^{Θ} = x kJ mol^{-1}

(ii) C(graphite) + $\frac{1}{2}O_2(g) \rightarrow CO(g)$; ΔrH^{Θ} = y kJ mol^{-1}

(iii) $CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$; ΔrH^{Θ} = z kJ mol^{-1}

Based on the above thermochemical equations, find out which one of the following algebraic relationships is correct?

(A) z = x + y **(B)** x = y + z **(C)** x = y – z **(D)** y = 2z – x

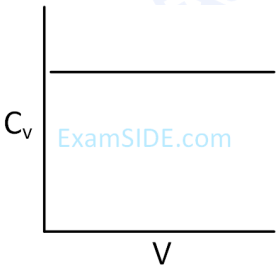
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Q172. JEE Main 2019 (Online) 12th January Morning Slot

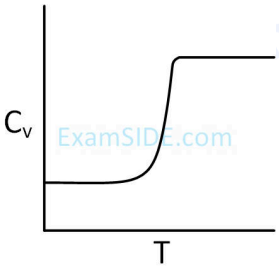
For diatomic ideal gas in a closed system, which of the following plots does not correctly describe the relation between various thermodynamic quantities?

Open solution

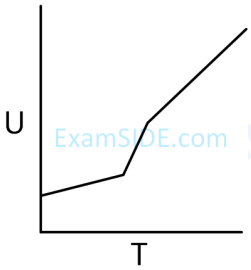
(A)



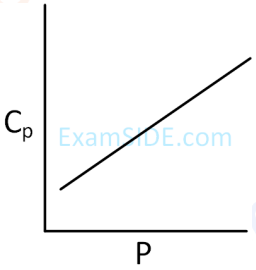
(B)



(C)



(D)



Q173. JEE Main 2019 (Online) 11th January Evening Slot

For the equilibrium,

$2H_2O \rightleftharpoons H_3O^+ + OH^-$, the value of ΔG^o at 298 K is approximately :

Open solution

(A) -80 kJ mol^{-1} (B) 100 kJ mol^{-1} (C) -100 kJ mol^{-1} (D) 80 kJ mol^{-1}

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Q174. JEE Main 2019 (Online) 11th January Evening Slot

The reaction, MgO(s) + C(s) $\rightarrow$ Mg(s) + CO(g), for which $\Delta_r H^o$ = 491.1 kJ mol^{-1} and $\Delta_r S^o$ = 198.0 JK^{-1} mol^{-1}, is not feasible at 298 K. Temperature above which reaction will be feasible is :

(A) 2480.3 K (B) 2040.5 K (C) 2380.5 K (D) 1890.0 K

Open solution

Q175. JEE Main 2019 (Online) 11th January Evening Slot

The standard reaction Gibbs energy for a chemical reaction at an absolute temperature T is given by $\Delta_r G^o = A - BT$

Where A and B are non-zero constants. Which of the following is TRUE about this reaction?

(A) Exothermic if B < 0
(B) Endothermic if A > 0
(C) Exothermic if A > 0 and B < 0
(D) Endothermic if A < 0 and B > 0

Open solution

Q176. JEE Main 2019 (Online) 11th January Morning Slot

For the chemical reaction X $\rightleftharpoons$ Y, the standard reaction Gibbs energy depends on temperature T (in K) as $\Delta_r G^o$ (in kJ mol^{-1}) = $120 - \frac{3}{8}$ T.

The major component of the reaction mixture at T is :

(A) Y if T = 300 K (B) Y if T = 280 K (C) X if T = 350 K (D) X if T = 315 K

Open solution

Q177. JEE Main 2019 (Online) 11th January Morning Slot

Two blocks of the same metal having same mass and at temperature T_1 and T_2, respectively, are brought in contact with each other and allowed to attain thermal equilibrium at constant pressure. The change in entropy, ΔS, for this process is :

(A) $2C_p \ln\left[\frac{(T_1+T_2)^{\frac{1}{2}}}{T_1T_2}\right]$

(B) $2C_p \ln\left[\frac{(T_1+T_2)}{2T_1T_2}\right]$

(C) $C_p \ln\left[\frac{(T_1+T_2)^2}{4T_1T_2}\right]$

(D) $2C_p \ln\left[\frac{(T_1+T_2)}{4T_1T_2}\right]$

Open solution

Q178. JEE Main 2019 (Online) 10th January Evening Slot

The process with negative entropy change is :

(A) Dissociation of $CaSO_4$ (s) to CaO(s) and SO_3(g)

(B) Dissolution of iodine in water

(C) Synthesis of ammonia from N_2 and H_2

(D) Sublimation of dry ice

Open solution

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Q179. JEE Main 2019 (Online) 10th January Evening Slot

An ideal gas undergoes isothermal compression from $5m^3$ to 1 m^3 against a constant external pressure of 4 Nm^{-2}. Heat released in this process is used to increase the temperature of 1 mole of Al. If molar heat capacity of Al is 24 J mol^{-1} K^{-1}, the temperature of Al increases by :

Open solution

(A) $\frac{2}{3}K$ (B) $\frac{3}{2}K$ (C) 1 K (D) 2 K

Q180. JEE Main 2019 (Online) 10th January Morning Slot

A process has ΔH = 200 J mol^{-1} and ΔS = 40 JK^{-1} mol^{-1}. Out of the values given below, choose the minimum temperature above which the process will be spontaneous :

Open solution

(A) 4 K (B) 20 K (C) 5 K (D) 12 K

Q181. JEE Main 2019 (Online) 9th January Evening Slot

The entropy change associated with the conversion of 1 kg of ice at 273 K to water vapours at 383 K is :

Open solution

(Specific heat of water liquid and water vapour are 4.2 kJ K^{-1} kg^{-1} and 2.0 kJ K^{-1} kg^{-1}; heat of liquid fusion and vapourisation of water are 334 kJ kg^{-1} and 2491 kJ kg^{-1}, respectively). (log 273 = 2.436, log 373 = 2.572, log 383 = 2.583)

(A) 7.90 kJ kg^{-1} K^{-1} (B) 2.64 kJ kg^{-1} K^{-1}

(C) 8.49 kJ kg^{-1} K^{-1} (D) 9.26 kJ kg^{-1} K^{-1}

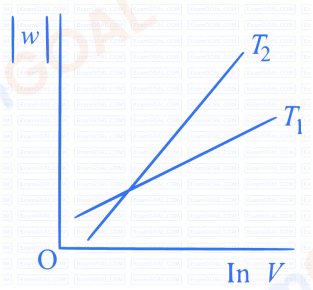
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Q182. JEE Main 2019 (Online) 9th January Morning Slot

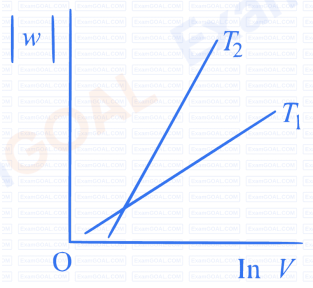
Open solution

Consider the reversible isothermal expansion of an ideal gas in a closed system at two different temperatures T_1 and T_2 ($T_1 < T_2$). The correct graphical depiction of the dependence of work done (w) on the final volume (V) is :

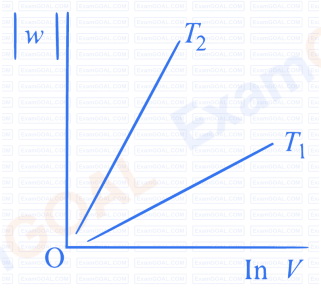
(A)



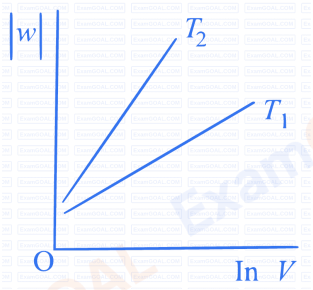
(B)



(C)



(D)



Q183. JEE Main 2018 (Online) 16th April Morning Slot

Open solution

At 320 K, a gas A_2 is 20% dissociated to A(g). The standard free energy change at 320 K and 1 atm in J mol^{-1} is approximately : (R = 8.314 JK^{-1} mol^{-1}; ln2 = 0.693; ln 3 = 1.098)

(A) 4763 (B) 2068 (C) 1844 (D) 4281

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Q184. JEE Main 2018 (Online) 16th April Morning Slot

For which of the following processes, ΔS is negative ?

(A) $H_2(g) \rightarrow 2H(g)$

(B) $N_2(g, 1\ atm) \rightarrow N_2(g, 5\ atm)$

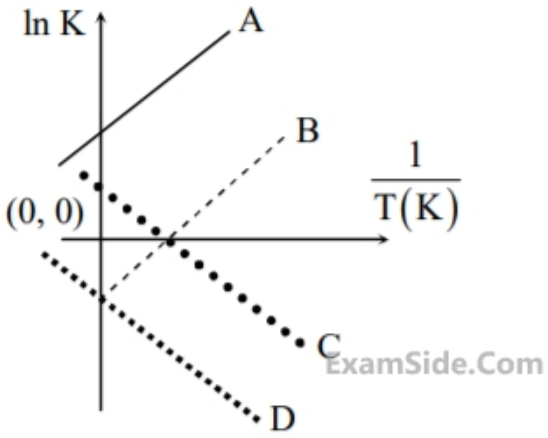
(C) C(diamond) $\rightarrow$ C(graphite)

(D) $N_2(g, 273\ K) \rightarrow N_2(g, 300\ K)$

Open solution

Q185. JEE Main 2018 (Offline)

Which of the following lines correctly show the temperature dependence of equilibrium constant K, for an exothermic reaction?



(A) A and D

(B) A and B

(C) B and C

(D) C and D

Open solution

Q186. JEE Main 2018 (Offline)

The combustion of benzene(l) gives $CO_2(g)$ and $H_2O(l)$. Given that heat of combustion of benzene at constant volume is -3263.9 kJ mol^{-1} at 25°C; heat of combustion (in kJ mol^{-1}) of benzene at constant pressure will be :

($R = 8.314\ JK^{-1}\ mol^{-1}$)

(A) −3267.6

(B) 4152.6

(C) −452.46

(D) 3260

Open solution

Q187. JEE Main 2018 (Online) 15th April Evening Slot

$\Delta_f G^o$ at 500 K for substance 'S' in liquid state and gaseous state are +100.7 kcal mol^{-1} and +103 kcal mol^{-1}, respectively. Vapour pressure of liquid 'S' at 500 K is approximately equal to : (R = 2 cal $K^{-1}\ mol^{-1}$)

(A) 0.1 atm

(B) 1 atm

(C) 10 atm

(D) 100 atm

Open solution

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Q188. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

Given

(i) $2Fe_2O_3(s) \rightarrow 4Fe(s) + 3O_2(g)$;

$\Delta_r G^o = +1487.0\ kJ\ mol^{-1}$

(ii) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$;

$\Delta_r G^o = -514.4\ kJ\ mol^{-1}$

Free energy change, $\Delta_r G^o$ for the reaction

$2Fe_2O_3(s) + 6CO(g) \rightarrow 4Fe(s) + 6CO_2(g)$ will be :

(A) $-112.4\ kJ\ mol^{-1}$ **(B)** $-56.2\ kJ\ mol^{-1}$ **(C)** $-168.2\ kJ\ mol^{-1}$ **(D)** $-208.0\ kJ\ mol^{-1}$

Q189. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

For which of the following reactions, ΔH is equal to ΔU?

(A) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

(B) $2HI(g) \rightarrow H_2(g) + I_2(g)$

(C) $2NO_2(g) \rightarrow N_2O_4(g)$

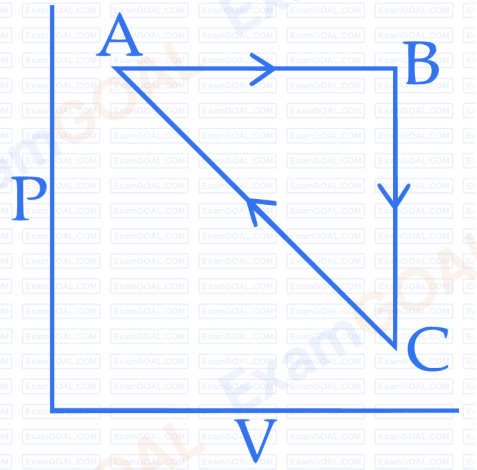
(D) $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

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Q190. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

An ideal gas undergoes a cyclic process as shown in Figure.



$\Delta U_{BC} = -5$ kJ mol^{-1}, $q_{AB} = 2$ kJ mol^{-1}, $W_{AB} = -5$ kJ mol^{-1}, $W_{CA} = 3$ kJ mol^{-1}. Heat absorbed by the system during process CA is :

(A) -5 kJ mol^{-1} (B) $+5$ kJ mol^{-1} (C) 18 kJ mol^{-1} (D) -18 kJ mol^{-1}

Q191. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

An ideal gas undergoes isothermal expansion at constant pressure. During the process :

(A) enthalpy increases but entropy decreases.

(B) enthalpy remains constant but entropy increases.

(C) enthalpy decreases but entropy increases.

(D) Both enthalpy and entropy remain constant.

Q192. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

A gas undergoes change from state A to state B. In this process, the heat absorbed and work done by the gas is 5 J and 8 J, respectively. Now gas is brought back to A by another process during which 3 J of heat is evolved. In this reverse process of B to A :

(A) 10 J of the work will be done by the gas.

(B) 6 J of the work will be done by the gas.

(C) 10 J of the work will be done by the surrounding on gas.

(D) 6 J of the work will be done by the surrounding on gas.

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Q193. JEE Main 2017 (Online) 8th April Morning Slot

For a reaction, A(g) $\rightarrow$ A(ℓ); $\Delta H = -3RT$.
The **correct** statement for the reaction is :

(A) $\Delta H = \Delta U \neq 0$

(B) $\Delta H = \Delta U = 0$

(C) $|\Delta H| < |\Delta U|$

(D) $|\Delta H| > |\Delta U|$

Open solution

Q194. JEE Main 2017 (Online) 8th April Morning Slot

The enthalpy change on freezing of 1 mol of water at 5°C to ice at −5°C is :

(Given $\Delta_{fus}H$ = 6 kJ mol^{-1} at 0°C,

$C_p(H_2O, \ell)$ = 75.3 J mol^{-1} K^{-1})

$C_p(H_2O, s)$ = 36.8 J mol^{-1} K^{-1})

(A) 5.44 kJ mol^{-1} (B) 5.81 kJ mol^{-1} (C) 6.56 kJ mol^{-1} (D) 6.00 kJ mol^{-1}

Open solution

Q195. JEE Main 2017 (Offline)

Given, $C_{(graphite)} + O_2 \rightarrow CO_2(g)$;

$\Delta_r H^o$ = - 393.5 kJ mol^{-1}

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$

$\Delta_r H^o$ = - 285.8 kJ mol^{-1}

$CO_2(g) + 2H_2O(l) \rightarrow CH_4(g) + 2O_2(g)$

$\Delta_r H^o$ = + 890.3 kJ mol^{-1}

Based on the above thermochemical equations, the value of $\Delta_r H^o$ at 298 K for the reaction

$C_{(graphite)} + 2H_2(g) \rightarrow CH_4(g)$ will be :

(A) +144.0 kJ mol^{-1} (B) − 74.8 kJ mol^{-1} (C) -144.0 kJ mol^{-1} (D) + 74.8 kJ mol^{-1}

Open solution

Q196. JEE Main 2017 (Offline)

ΔU is equal to :

(A) Isobaric work (B) Adiabatic work (C) Isothermal work (D) Isochoric work

Open solution

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Q197. JEE Main 2016 (Online) 10th April Morning Slot

If 100 mole of H_2O_2 decompose at 1 bar and 300 K, the work done (kJ) by one mole of $O_2(g)$ as it expands against 1 bar pressure is :

Open solution

$2H_2O_2(l) \rightleftharpoons 2H_2O(l) + O_2(g)$

$(R = 8.3\ J\ K^{-1}\ mol^{-1})$

(A) 62.25 **(B)** 124.50 **(C)** 249.00 **(D)** 498.00

Q198. JEE Main 2016 (Online) 9th April Morning Slot

A reaction at 1 bar is non-spontaneous at low temperature but becomes spontaneous at high temperature. Identify the correct statement about the reaction among the following :

Open solution

(A) Both ΔH and ΔS are negative.

(B) Both ΔH and ΔS are positive.

(C) ΔH is positive while ΔS is negative.

(D) ΔH is negative while ΔS is positive.

Q199. JEE Main 2016 (Online) 9th April Morning Slot

For the reaction,
$A(g) + B(g) \rightarrow C(g) + D(g)$, ΔH^o and ΔS^o are, respectively, $-29.8\ kJ\ mol^{-1}$ and $-0.100\ kJ\ K^{-1}\ mol^{-1}$ at 298 K. The equilibrium constant for the reaction at 298 K is :

Open solution

(A) 1.0×10^{-10} **(B)** 1.0×10^{10} **(C)** 10 **(D)** 1

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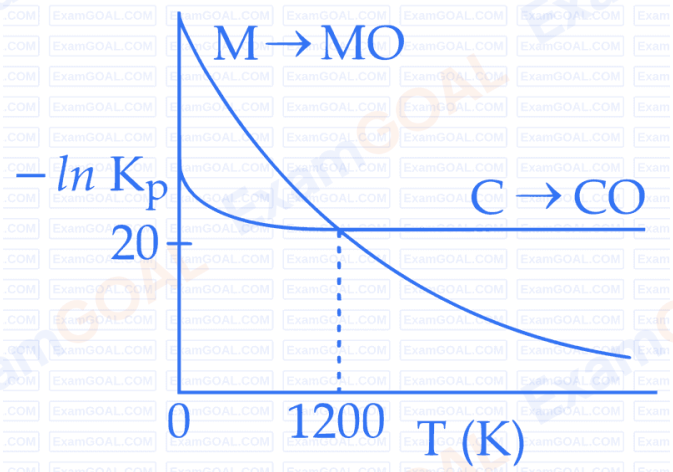
Q200. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

The plot shows the variation of $-ln\ K_p$ versus temperature for the two reactions.

$M(s) + \frac{1}{2} O_2(g) \rightarrow MO(s)$ and

$C(s) + \frac{1}{2} O_2(g) \rightarrow CO(s)$



Identify the correct statement :

(A) At T > 1200 K, carbon will reduce MO(s) to M(s).

(B) At T < 1200 K, the reaction

$MO(s) + C(s) \rightarrow M(s) + CO(g)$ is spontaneous.

(C) At T < 1200 K, oxidation of carbon is unfavourable

(D) Oxidation of carbon is favourable at all temperatures.

Q201. JEE Main 2016 (Offline)

Open solution

The heats of combustion of carbon and carbon monoxide are −393.5 and −283.5 kJ mol^{-1}, respectively. The heat of formation (in kJ) of carbon monoxide per mole is :

(A) 676.5 **(B)** -676.5 **(C)** −110 **(D)** 110.5

Q202. JEE Main 2015 (Online) 10th April Morning Slot

Open solution

The heat of atomization of methane and ethane are 360 kJ/mol and 620 kJ/mol, respectively. The longest wavelength of light capable of breaking the C - C bond is (Avogadro number = 6.02×10^{23}, h = 6.62×10^{-34} J s) :

(A) 1.49×10^3 nm **(B)** 2.48×10^3 nm **(C)** 2.48×10^4 nm **(D)** 1.49×10^4 nm

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Q203. JEE Main 2015 (Offline)

Open solution

The following reaction is performed at 298 K
$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$
The standard free energy of formation of NO(g) is 86.6 kJ/mol at 298 K. What is the standard free energy of formation of $NO_2(g)$ at 298 K? ($K_P = 1.6 \times 10^{12}$)

(A) $86600 + R(298)\ln(1.6 \times 10^{12})$
(B) $86600 - \frac{ln(1.6\times10^{12})}{R(298)}$
(C) $0.5[2\times86{,}600 - R(298)\ln(1.6\times10^{12})]$
(D) $R(298)\ln(1.6\times10^{12}) - 86600$

Q204. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

The entropy (S^0) of the following substances are :
CH_4 (g) 186.2 J K^{-1} mol^{-1}
O_2 (g) 205.0 J K^{-1} mol^{-1}
CO_2 (g) 213.6 J K^{-1} mol^{-1}
H_2O (l) 69.9 J K^{-1} mol^{-1}

The entropy change (ΔS^o) for the reaction
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ is :

(A) -312.5 J K^{-1} mol^{-1}
(B) -242.8 J K^{-1} mol^{-1}
(C) -108.1 J K^{-1} mol^{-1}
(D) -37.6 J K^{-1} mol^{-1}

Q205. JEE Main 2014 (Online) 12th April Morning Slot

Open solution

The standard enthalpy of formation ($\Delta_f H^o_{298}$) for methane, CH_4 is -74.9 kJ mol^{-1}. In order to calculate the average energy given out in the formation of a C – H bond from this it is necessary to know which one of the following ?

(A) the dissociation energy of the hydrogen molecule, H_2.
(B) the first four ionisation energies of carbon.
(C) the dissociation energy of H_2 and enthalpy of sublimation of carbon (graphite) :
(D) the first four ionisation energies of carbon and electron affinity of hydrogen.

Q206. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

The molar heat capacity (C_p) of CD_2O is 10 cals at 1000 K. The change in entropy associated with cooling of 32 g of CD_2O vapour from 1000 K to 100 K at constant pressure will be :
(D = deuterium, at. mass = 2 u)

(A) 23.03 cal deg–1 (B) -23.03 cal deg^{-1} (C) 2.303 cal deg^{-1} (D) -2.303 cal deg^{-1}

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Q207. JEE Main 2014 (Online) 9th April Morning Slot

The standard enthalpy of formation of NH_3 is $-$ 46.0 kJ/mol. If the enthalpy of formation of H_2 from its atoms is $-$ 436 kJ/mol and that of N_2 is $-$ 712 kJ/mol, the average bond enthalpy of N $-$ H bond in NH_3 is :

Open solution

(A) $-$ 1102 kJ/mol (B) $-$ 964 kJ/mol (C) + 352 kJ/mol (D) + 1056 kJ/mol

Q208. JEE Main 2014 (Offline)

For complete combustion of ethanol, $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$ the amount of heat produced as measured in bomb calorimeter, is 1364.47 kJ mol^{-1} at 25°C. Assuming ideality the Enthalpy of combustion, $\Delta_C H$, for the reaction will be : (R = 8.314 J K^{-1} mol^{-1})

Open solution

(A) -1460.50 kJ mol^{-1} (B) $-$ 1350.50 kJ mol^{-1}

(C) $-$ 1366.95 kJ mol^{-1} (D) $-$ 1361.95 kJ mol^{-1}

Q209. JEE Main 2013 (Online) 25th April Morning Slot

Given that: (i) $\Delta_f H^\circ$ of N_2O is 82 kJ mol^{-1} (ii) Bond energies of $N \equiv N$, $N = N$, $O = O$ and $N = O$ are 946, 418, 498 and 607 kJ mol^{-1} respectively. The resonance energy of N_2O is :

Open solution

(A) -88 kJ (B) -66 kJ (C) -62 kJ (D) -44 kJ

Q210. JEE Main 2013 (Online) 23th April Morning Slot

Which of the following statements/relationships is not correct in thermodynamic changes?

Open solution

(A) $\Delta U = 0$ (isothermal reversible expansion of a gas)

(B) $w = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

(C) $w = nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

(D) For a system of constant volume heat involved directly changes to internal energy.

Q211. JEE Main 2013 (Online) 22th April Morning Slot

Given Reaction Energy Change (in kJ):

Li(s) $\rightarrow$ Li(g) : 161

Li(g) $\rightarrow$ Li^+(g) : 520

$\frac{1}{2}F_2(g) \rightarrow$ F(g) : 77

F(g) + e^- $\rightarrow$ F^-(g) : (Electron gain enthalpy)

Li^+(g) + F^-(g) $\rightarrow$ LiF(s) : -1047

Li(s) + $\frac{1}{2}F_2$(g) $\rightarrow$ LiF(s) : -617

Based on data provided, the value of electron gain enthalpy of fluorine would be :

Open solution

(A) -300 kJ mol^{-1} (B) -350 kJ mol^{-1} (C) -328 kJ mol^{-1} (D) -228 kJ mol^{-1}

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Q212. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

Given: (I) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$; $\Delta H^\circ{}_{298K} = -285.9$ kJ mol^{-1}
(II) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$; $\Delta H^\circ{}_{298K} = -241.8$ kJ mol^{-1}
The molar enthalpy of vapourisation of water will be :

(A) 241.8 kJ mol^{-1} (B) 22.0 kJ mol^{-1} (C) 44.1 kJ mol^{-1} (D) 527.7 kJ mol^{-1}

Q213. JEE Main 2013 (Offline)

Open solution

A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of 37.0^{o}C. As it does so, it absorbs 208J of heat. The values of q and w for the process will be :
(R = 8.314 J/mol K) (ln 7.5 = 2.01)

(A) q = – 208 J, w = – 208 J (B) q = – 208 J, w = + 208 J
(C) q = + 208 J, w = + 208 J (D) q = + 208 J, w = – 208 J

Q214. AIEEE 2012 (Online) 26th May

Open solution

One mole of an ideal gas is expanded isothermally and reversibly to half of its initial pressure. ΔS for the process in $\mathrm{J\,K^{-1}\,mol^{-1}}$ is [$\ln 2 = 0.693$ and $R = 8.314\ \mathrm{J/(mol\,K)}$]

(A) 6.76 (B) 5.76 (C) 10.76 (D) 8.03

Q215. AIEEE 2012 (Online) 19th May

Open solution

The enthalpy of neutralisation of $\mathrm{NH_4OH}$ with HCl is $-51.46\ \mathrm{kJ\ mol^{-1}}$ and the enthalpy of neutralisation of NaOH with HCl is $-55.90\ \mathrm{kJ\ mol^{-1}}$. The enthalpy of ionisation of $\mathrm{NH_4OH}$ is

(A) $-107.36\ \mathrm{kJ\ mol^{-1}}$ (B) $-4.44\ \mathrm{kJ\ mol^{-1}}$ (C) $+107.36\ \mathrm{kJ\ mol^{-1}}$ (D) $+4.44\ \mathrm{kJ\ mol^{-1}}$

Q216. AIEEE 2012 (Online) 12th May

Open solution

The difference between the reaction enthalpy change $(\Delta_r H)$ and reaction internal energy change $(\Delta_r U)$ for the reaction:
$\mathrm{2C_6H_6(l) + 15O_2(g) \longrightarrow 12CO_2(g) + 6H_2O(l)}$
at 300 K is $(R = 8.314\ \mathrm{J\,mol^{-1}\,K^{-1}})$

(A) $0\ \mathrm{J\,mol^{-1}}$ (B) $2490\ \mathrm{J\,mol^{-1}}$ (C) $-2490\ \mathrm{J\,mol^{-1}}$ (D) $-7482\ \mathrm{J\,mol^{-1}}$

Q217. AIEEE 2012 (Online) 7th May

Open solution

The entropy of a sample of a certain substance increases by $0.836\ \mathrm{J\,K^{-1}}$ on adding reversibly 0.3344 J of heat at constant temperature. The temperature of the sample is:

(A) 2.5 K (B) 0.3 K (C) 0.016 K (D) 0.4 K

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Q218. AIEEE 2012

The incorrect expression among the following is :

(A) $\frac{\Delta G_{system}}{\Delta S_{total}} = -T$

(B) In isothermal process $w_{reversible} = -nRT \ln \frac{V_f}{V_i}$

(C) In $K = \frac{\Delta H^o - T\Delta S^o}{RT}$

(D) $K = e^{-\Delta G^o/RT}$

Open solution

Q219. AIEEE 2011

The entropy change involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm^3 to a volume of 100 dm^3 at 27°C is :

(A) 35.8 J mol^{-1} K^{-1} (B) 32.3 J mol^{-1} K^{-1} (C) 42.3 J mol^{-1} K^{-1} (D) 38.3 J mol^{-1} K^{-1}

Open solution

Q220. AIEEE 2010

The standard enthalpy of formation of NH_3 is −46.0 kJ mol^{-1}. If the enthalpy of formation of H_2 from its atoms is −436 kJ mol^{-1} and that of N_2 is −712 kJ mol^{-1}, the average bond enthalpy of N–H bond in NH_3 is :

(A) −964 kJ mol^{-1} (B) +352 kJ mol^{-1} (C) + 1056 kJ mol^{-1} (D) −1102 kJ mol^{-1}

Open solution

Q221. AIEEE 2010

For a particular reversible reaction at temperature T, ΔH and ΔS were found to be both +ve. If T_e is the temperature at equilibrium, the reaction would be spontaneous when :

(A) $T_e > T$ (B) $T > T_e$ (C) T_e is 5 times T (D) $T = T_e$

Open solution

Q222. AIEEE 2009

On the basis of the following thermochemical data :
($\Delta_f G^o H^+_{(aq)}$ = 0)

$H_2O(l) \rightarrow H^+(aq) + OH^-(aq)$; ΔH = 57.32 kJ

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$; ΔH = -286.20 kJ

The value of enthalpy of formation of OH^- ion at 25°C is :

(A) -22.88 kJ (B) -228.88 kJ (C) +228.88 kJ (D) -343.52 kJ

Open solution

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Q223. AIEEE 2008

Open solution

Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:

$\frac{1}{2}Cl_2(g) \xrightarrow{\frac{1}{2}\Delta_{diss}H^{\Theta}} Cl(g) \xrightarrow{\Delta_{eg}H^{\Theta}} Cl^-(g) \xrightarrow{\Delta_{Hyd}H^{\Theta}} Cl^-(aq)$

The energy involved in the conversion of $\frac{1}{2}Cl_2(g)$ to $Cl^-(aq)$ (using the data, $\Delta_{diss}H^{\Theta}_{Cl_2}$ = 240 kJ/mol, $\Delta_{eg}H^{\Theta}_{Cl}$ = -349 kJ/mol, $\Delta_{hyd}H^{\Theta}_{Cl^-}$ = - 381 kJ/mol) will be :

(A) +152 kJ mol^{-1} (B) −610 kJ mol^{-1} (C) −850 kJ mol^{-1} (D) +120 kJ mol^{-1}

Q224. AIEEE 2008

Open solution

Standard entropy of X_2, Y_2 and XY_3 are 60, 40 and 50 JK^{-1} mol^{-1} , respectively. For the reaction, $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, ΔH = -30 kJ, to be at equilibrium, the temperature will be :

(A) 1250 K (B) 500 K (C) 750 K (D) 1000 K

Q225. AIEEE 2007

Open solution

In conversion of lime-stone to lime,
$CaCO_3(s) \rightarrow CaO(s) + CO_2$ (g) the vales of ΔH° and ΔS° are +179.1 kJ mol−1 and 160.2 J/K respectively at 298 K and 1 bar. Assuming that ΔH° do not change with temperature, temperature above which conversion of limestone to lime will be spontaneous is :

(A) 1008 K (B) 1200 (C) 845 K (D) 1118 K

Q226. AIEEE 2007

Open solution

Identify the correct statement regarding a spontaneous process :

(A) For a spontaneous process in an isolated system, the change in entropy is positive

(B) Endothermic processes are never spontaneous

(C) Exothermic processes are always spontaneous

(D) Lowering of energy in the reaction process is the only criterion for spontaneity

Q227. AIEEE 2007

Open solution

Assuming that water vapour is an ideal gas, the internal energy change (ΔU) when 1 mol of water is vapourised at 1 bar pressure and $100^{\circ}C$ (Given : molar enthalpy of vapourisation of water at 1 bar and $373\ K = 41\ kJ\ mol^{-1}$
and $R = 8.3\ J\ mol^{-1}\ K^{-1}$) will be

(A) $41.00\ kJ\ mol^{-1}$ (B) $4.100\ kJ\ mol^{-1}$ (C) $3.7904\ kJ\ mol^{-1}$ (D) $37.904\ kJ\ mol^{-1}$

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Q228. AIEEE 2006

An ideal gas is allowed to expand both reversibly and irreversibly in an isolated system. If T_i is the initial temperature and T_f is the final temperature, which of the following statements is correct?

Open solution

(A) $(T_f)_{irrev} > (T_f)_{rev}$

(B) $(T_f)_{rev} = (T_f)_{irrev}$

(C) $T_f > T_i$ for reversible process but $T_f = T_i$ for irreversible process

(D) $T_f = T_i$ for both reversible and irreversible processes

Q229. AIEEE 2006

The standard enthalpy of formation $\Delta_f H^o$ at 298 K for methane, $CH_4(g)$, is -74.8 kJ mol^{-1}. The additional information required to determine the average energy for C – H bond formation would be :

Open solution

(A) the dissociation energy of H_2 and enthalpy of sublimation of carbon

(B) latent heat of vapourization of methane

(C) the first four ionization energies of carbon and electron gain enthalpy of hydrogen

(D) the dissociation energy of hydrogen molecule, H_2

Q230. AIEEE 2006

The enthalpy changes for the following processes are listed below :

Open solution

$Cl_2(g) = 2Cl(g)$, 242.3 kJ mol^{-1}

$I_2(g) = 2I(g)$, 151.0 kJ mol^{-1}

$ICl(g) = I(g) + Cl(g)$, 211.3 kJ mol^{-1}

$I_2(s) = I_2(g)$, 62.76 kJ mol^{-1}

Given that the standard states for iodine and chlorine are $I_2(s)$ and $Cl_2(g)$, the standard enthalpy of formation for $ICl(g)$ is :

(A) -14.6 kJ mol^{-1} (B) -16.8 kJ mol^{-1} (C) $+16.8$ kJ mol^{-1} (D) $+244.8$ kJ mol^{-1}

Q231. AIEEE 2006

$(\Delta H - \Delta U)$ for the formation of carbon monoxide (CO) from its elements at 298 K is : (R = 8.314 J K^{-1} mol^{-1})

Open solution

(A) -1238.78 J mol^{-1} (B) 1238.78 J mol^{-1} (C) -2477.57 J mol^{-1} (D) 2477.57 J mol^{-1}

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Q232. AIEEE 2005

Consider the reaction: $N_2 + 3H_2 \rightarrow 2NH_3$ carried out at constant temperature and pressure. If ΔH and ΔU are the enthalpy and internal energy changes for the reaction, which of the following expressions is true?

Open solution

(A) $\Delta H > \Delta U$ (B) $\Delta H < \Delta U$ (C) $\Delta H = \Delta U$ (D) $\Delta H = 0$

Q233. AIEEE 2005

If the bond dissociation energies of XY, X_2 and Y_2 (all diatomic molecules) are in the ratio of 1:1:0.5 and ΔH_f for the formation of XY is -200 kJ $mole^{-1}$. The bond dissociation energy of X_2 will be :

Open solution

(A) 100 kJ mol^{-1} (B) 200 kJ mol^{-1} (C) 300 kJ mol^{-1} (D) 800 kJ mol^{-1}

Q234. AIEEE 2004

An ideal gas expands in volume from 1×10^{-3} m^3 to 1×10^{-2} m^3 at 300 K against a constant pressure of 1×10^5 Nm^{-2}. The work done is :

Open solution

(A) -900 J (B) 900 kJ (C) 270 kJ (D) -900 kJ

Q235. AIEEE 2004

The enthalpies of combustion of carbon and carbon monoxide are -393.5 and -283 kJ mol^{-1} respectively. The enthalpy of formation of carbon monoxide per mole is :

Open solution

(A) 110.5 kJ (B) -110.5 kJ (C) -676.5 kJ (D) 676.5 kJ

Q236. AIEEE 2003

The internal energy change when a system goes from state A to B is 40 kJ/mole. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy?

Open solution

(A) > 40 kJ (B) < 40 kJ (C) Zero (D) 40 kJ

Q237. AIEEE 2003

If at 298 K the bond energies of C - H, C - C, C = C and H - H bonds are respectively 414, 347, 615 and 435 kJ/mol, the value of enthalpy change for the reaction
$H_2C = CH_2(g) + H_2(g) \rightarrow H_3C - CH_3(g)$ at 298 K will be :

Open solution

(A) - 250 kJ (B) + 125 kJ (C) - 125 kJ (D) + 250 kJ

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Q238. AIEEE 2003

In an irreversible process taking place at constant T and P and in which only pressure-volume work is being done, the change in Gibbs free energy (dG) and change in entropy (dS), satisfy the criteria :

(A) $(dS)_{V, E} > 0, (dG)_{T, P} < 0$

(B) $(dS)_{V, E} = 0, (dG)_{T, P} = 0$

(C) $(dS)_{V, E} = 0, (dG)_{T, P} > 0$

(D) $(dS)_{V, E} < 0, (dG)_{T, P} < 0$

Open solution

Q239. AIEEE 2003

The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K_c is :

(A) $-\Delta G = RT \ln K_c$

(B) $\Delta G^o = RT \ln K_c$

(C) $-\Delta G^o = RT \ln K_c$

(D) $\Delta G = RT \ln K_c$

Open solution

Q240. AIEEE 2003

The enthalpy change for a reaction does not depend upon :

(A) use of different reactants for the same product

(B) the nature of intermediate reaction steps

(C) the differences in initial or final temperature of involved substances

(D) the physical states of reactants and products

Open solution

Q241. AIEEE 2002

If an endothermic reaction is non-spontaneous at freezing point of water and becomes feasible at its boiling point, then :

(A) ΔH is -ve, ΔS is +ve

(B) ΔH and ΔS are both +ve

(C) ΔH and ΔS are both -ve

(D) ΔH is +ve, ΔS is -ve

Open solution

Q242. AIEEE 2002

A heat engine absorbs heat Q_1 at temperature T_1 and heat Q_2 at temperature T_2. Work done by the engine is J $(Q_1 + Q_2)$. This data :

(A) violates 1st law of thermodynamics

(B) violates 1st law of thermodynamics if Q_1 is +ve

(C) violates 1st law of thermodynamics if Q_1 is -ve

(D) does not violate 1st law of thermodynamics

Open solution

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Q243. AIEEE 2002

For the reactions
$C + O_2 \rightarrow CO_2$; ΔH = -393 J
$2Zn + O_2 \rightarrow 2ZnO$; ΔH = -412 J

(A) carbon can oxidise Zn
(B) oxidation of carbon is not feasible
(C) oxidation of Zn is not feasible
(D) Zn can oxidise carbon

Open solution

Q244. AIEEE 2002

The heat required to raise the temperature of body by 1 K is called :

(A) specific heat (B) thermal capacity (C) water equivalent (D) none of these

Open solution

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Answer Key

Numerical

1. **1126**	2. **34**	3. **3**	4. **727**	5. **1031**
6. **70**	7. **98**	8. **304**	9. **2500**	10. **25**
11. **48**	12. **466**	13. **700**	14. **200**	15. **2850**
16. **95**	17. **400**	18. **2**	19. **82**	20. **38**
21. **55**	22. **2000**	23. **274**	24. **6535**	25. **150**
26. **200**	27. **125**	28. **57**	29. **28721**	30. **163**
31. **492**	32. **200**	33. **56**	34. **37**	35. **1200**
36. **1070**	37. **400**	38. **3**	39. **4**	40. **4**
41. **4**	42. **1411**	43. **1200**	44. **278**	45. **2**
46. **10**	47. **1006**	48. **499**	49. **173**	50. **610**
51. **150**	52. **0**	53. **847**	54. **360**	55. **620**
56. **2**	57. **2**	58. **54**	59. **3**	60. **35**
61. **2**	62. **57**	63. **200**	64. **2**	65. **104**
66. **1718**	67. **195**	68. **117**	69. **600**	70. **0**
71. **8630**	72. **38**	73. **727**	74. **300**	75. **5**
76. **84**	77. **964**	78. **82**	79. **38**	80. **718**
81. **82**	82. **31**	83. **50**	84. **101**	85. **21**
86. **336**	87. **667**	88. **128**	89. **230**	90. **2218**
91. **309**	92. **200**	93. **15**	94. **741**	95. **5576**
96. **855**	97. **-13540 to -13537**	98. **326400**	99. **189494 to 189495**	100. **6.25**
101. **48**	102. **-192.5**	103. **-2.7**		

MCQ (Single Correct Answer)

104. **C**	105. **C**	106. **B**	107. **B**	108. **D**
109. **D**	110. **D**	111. **C**	112. **A**	113. **D**
114. **C**	115. **A**	116. **D**	117. **A**	118. **A**
119. **B**	120. **B**	121. **B**	122. **C**	123. **C**
124. **C**	125. **B**	126. **C**	127. **A**	128. **A**
129. **B**	130. **B**	131. **B**	132. **C**	133. **C**
134. **C**	135. **B**	136. **D**	137. **D**	138. **D**
139. **C**	140. **B**	141. **B**	142. **A**	143. **A**

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144. **B**	145. **A**	146. **B**	147. **A**	148. **A**
149. **A**	150. **A**	151. **B**	152. **C**	153. **C**
154. **B**	155. **C**	156. **B**	157. **A**	158. **A**
159. **C**	160. **C**	161. **B**	162. **B**	163. **A**
164. **A**	165. **B**	166. **A**	167. **A**	168. **A**
169. **A**	170. **B**	171. **B**	172. **D**	173. **D**
174. **A**	175. **B**	176. **D**	177. **C**	178. **C**
179. **A**	180. **C**	181. **D**	182. **B**	183. **D**
184. **B**	185. **B**	186. **A**	187. **A**	188. **B**
189. **B**	190. **B**	191. **B**	192. **D**	193. **D**
194. **C**	195. **B**	196. **B**	197. **C**	198. **B**
199. **D**	200. **B**	201. **C**	202. **A**	203. **C**
204. **B**	205. **C**	206. **B**	207. **C**	208. **C**
209. **A**	210. **C**	211. **C**	212. **C**	213. **D**
214. **B**	215. **D**	216. **D**	217. **D**	218. **C**
219. **D**	220. **B**	221. **B**	222. **B**	223. **B**
224. **C**	225. **D**	226. **A**	227. **D**	228. **A**
229. **A**	230. **C**	231. **B**	232. **B**	233. **D**
234. **A**	235. **B**	236. **C**	237. **C**	238. **A**
239. **C**	240. **B**	241. **B**	242. **A**	243. **D**
244. **B**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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